INCH-POUND

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w/CHANGE 2
6 Oct 2016
SUPERSEDING
MIL-STD-1669A
w/CHANGE 1
1 April 2011

DEPARTMENT OF DEFENSE
INTERFACE STANDARD

INSERT ARRANGEMENTS
FOR
MIL-DTL-26482

ENVIRONMENT RESISTING, CIRCULAR
ELECTRICAL CONNECTORS

AMSC N/A

FSC 5935

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FOREWORD

Insert Arrangements for MIL-DTL-26482 Environment Resisting, Circular, Electrical Connectors.

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Comments, suggestions, or questions on this document should be addressed to: Commander, DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990, or emailed to CircularConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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SUMMARY OF CHANGE 2 MODIFICATIONS

1. Editorial changes to “Foreword” on page ii: These changes include an update to the preamble in accordance with MIL-STD-962, and an update the website link for the Acquisition Streamlining and Standardization Information System (ASSIST).
2. Editorial changes to section 2, “Applicable Documents”, pages 1-2: These changes include updates, when applicable, to the parenthetical source statements below the listed documents, and updates to document numbers or document titles, when applicable.
3. Editorial changes to section 6 on pages 5 and 6: These changes include an update to the ASSIST website link referenced in paragraph 6.3, and the addition of a new Table III, “Cross-reference of insert arrangements by shell size”. The new table is intended to provide the user with hyperlinks to each insert arrangement in the standard.
4. Editorial correction to figure 8, insert arrangement 20-39, page 40, contact cavity location “p”: Change Y-axis location from “+.189” to “-.189”.
5. Marking requirement changes to add tracer lines to the following insert arrangements: 20-41, 22-34, 24-27, 24-31 and 24-61. The tracer lines, consisting of parentheses around the first and every 10th contact cavity in the contact identification lettering sequence, is intended as an additional aid to users.
6. Editorial change to “Concluding Material”: This change includes an update to the ASSIST website link.

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CONCLUDING MATERIAL 63

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1. SCOPE

1.1 Scope. This standard covers insert arrangements for use with MIL-DTL-26482 environment resisting, circular, electrical connectors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-26482	-	Connectors, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles and Plugs, General Specification for

(Copies these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME Y14.5	-	Dimensioning and Tolerancing

(Copies of these documents are available online at http://www.asme.org.)

SAE INTERNATIONAL

SAE-AS39029/4	Contacts, Electrical Connector, Pin, Crimp Removable, (for MIL-DTL-26482 Series 2, AS81703 Series 3 and MIL-DTL-83723 Series I and III, and MIL-DTL-83733 Connectors)
SAE-AS39029/5	Contacts, Electrical Connector, Socket, Crimp Removable (for MIL-DTL-26482 Series 2, AS81703 Series 3 and MIL-DTL-83723 Series I and III, MIL-DTL-83733 Connectors and MIL-DTL-12883/40 and /41 Relay Sockets)

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SAE-AS39029/7	Contacts, Electrical Connector, Pin, Crimp Removable, Shielded (for MIL- C-26482 Series 2 and MIL-C-81703 Series 3 Connectors)
SAE-AS39029/8	Contacts, Electrical Connector, Socket, Crimp Removable, Shielded, (for MIL-C-26482 Series 2 and MIL-C-81703 Series 3 Connectors)
SAE-AS39029/9	Contacts, Electrical Connector, Pin, Crimp Removable, Thermocouple, (for MIL-DTL-26482 Series 2 and AS81703 Series 3, MIL-DTL-83723 Series I and III, and MIL-DTL-83733 Connectors)
SAE-AS39029/10	Contacts, Electrical Connector, Socket, Crimp Removable, Thermocouple, (for MIL-DTL-26482 Series 2 and AS81703 Series 3, MIL-DTL-83723 Series I and III, and MIL-DTL-83733 Connectors)
SAE-AS39029/22	Contacts, Electrical, Connector, Socket, Crimp Removable, (for AS81714 Terminal Junction System Series II)
SAE-AS39029/23	Contacts, Electrical Connector, Pin, Crimp Removable, Shielded, (for MIL-C-26482 Series 1 Connectors)
SAE-AS39029/24	Contacts, Electrical Connector, Socket, Crimp Removable, Shielded, (for MIL-C-26482 Series 1 Connectors)
SAE-AS39029/25	Contacts, Electrical Connectors, Pins, Crimp Removable, Shielded, (for MIL-DTL-26482 Series 1 Connectors)
SAE-AS39029/26	Contacts, Electrical Connector, Socket, Crimp Removable, Shielded, (for MIL-DTL-26482 Series 1 Connectors)
SAE-AS39029/31	Contacts, Electrical Connector, Pin, Crimp Removable, (for MIL-DTL-26482 Series 1, MIL-DTL-26500 and MIL-DTL-26518 Connectors)
SAE-AS39029/32	Contacts, Electrical Connector, Socket, Crimp Removable, (for MIL-DTL-26482 Series 1, MIL-DTL-26500 and MIL-DTL-26518 Connectors)

(Copies of these documents are available online at http://www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions. The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.

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4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economical advantageous life cycle costs.

5. DETAILED REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable section of this standard and the following dimensional data:

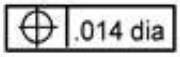
a. Dimensioning and tolerancing in accordance with ASME Y14.5. Dimensions are true position (TP) and are in inches.

b. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based on 1 inch = 25.4 mm.

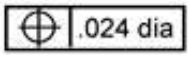
c. Dimensions and markings shown are for engaging face of pin insert; socket insert is opposite.

d. The following tolerances apply to an insert installed in the shell:

1. Center of each hole in the insert shall be located at TP within .014 in (0.356 mm) diameter.



2. Center of engaging end of each contact shall be located at TP within .024 in (0.610 mm) diameter.



e. Unless otherwise indicated, dimensions are symmetrical about centerline.

f. ↓ indicates ℄ (centerline) of insert arrangement.

5.2 Main key or keyway polarization.

a. Each insert arrangement is shown in the “normal position” in the shell.

b. The “master key or keyway” of the shell and the “indexing radius of the insert” are shown on figure 1.

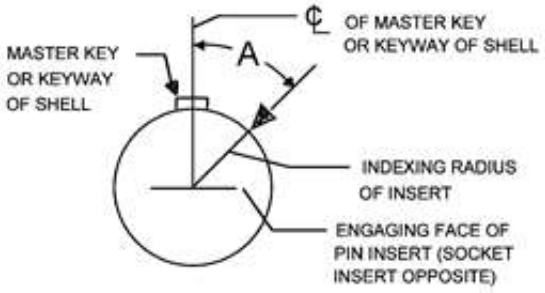


FIGURE 1. Insert position.

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c. In the “normal position” the indexing radius coincides with the ℄ of the “master key or keyway of the shell”.

d. In “alternate insert positions” the pin insert is rotated “A” degrees clockwise relative to ℄ of “master key or keyway of shell” as indicated.

e. In “alternate positions” the socket insert is rotated “A” degrees counter-clockwise relative to ℄ of “master key or keyway of shell”.

f. See applicable section of this standard for insert positions.

g. Four keys or keyways (maximum material condition) (MMC) and insert shall be located within .0035 inches (0.089 mm) either side of true position (TP) relative to master key or keyway (MMC) and shell outside diameter or inside diameter (MMC).

5.3 Contacts. Solder contacts shall be in accordance with MIL-DTL-26482. Crimp contacts shall be in accordance with applicable document as specified in table I.

TABLE I. Crimp contacts.

Contact size	Applicable documents	
	Series 1	Series 2
20 Power	SAE-AS39029/31, SAE-AS39029/32	SAE-AS39029/4, SAE-AS39029/5
16 Power	SAE-AS39029/31, SAE-AS39029/32	SAE-AS39029/4, SAE-AS39029/5
12 Power	SAE-AS39029/31, SAE-AS39029/32	SAE-AS39029/4, SAE-AS39029/5
12 Shielded	SAE-AS39029/25, SAE-AS39029/26	SAE-AS39029/7, SAE-AS39029/8
8 Shielded	SAE-AS39029/23, SAE-AS39029/24	-------
20 Thermocouple	-------	SAE-AS39029/9, SAE-AS39029/10
20 Hermetic (class N)	-------	SAE-AS39029/22
16 Hermetic (class N)	-------	SAE-AS39029/22

5.4 Marking. Marking shall be in accordance with MIL-DTL-26482 and as shown in the applicable section of this standard.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard covers insert arrangements for use with MIL-DTL-26482 environment resisting, circular, electrical connectors. Information pertaining to the shell size, arrangement number, number of contacts, service rating, contact location and superseding information is contained in this document.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

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6.3 International standardization agreement implementation. This standard implements NATO STANAG 7067 ED.2(2). When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the U.S National Point of Contact for the international standardization agreement, as identified in the ASSIST database at http://quicksearch.dla.mil/.

6.4 Cross-reference. See table II for cross-reference of previous sections and current figures. See table III for a cross-reference of figures to their associated insert arrangements for each shell size. For the convenience of the user, table III includes hyperlinks to each insert arrangement.

TABLE II. Cross-reference of figures to sections of previous revisions.

Title	Current figure	Section of previous revisions
Shell size 8	2	10
Shell size 10	3	20
Shell size 12	4	30
Shell size 14	5	40
Shell size 16	6	50
Shell size 18	7	60
Shell size 20	8	70
Shell size 22	9	80
Shell size 24	10	90

Table III. Cross-reference of insert arrangements by shell size.

Shell size 8, figure 2			
Insert arrangement	Page	Insert arrangement	Page
8-2	7	8-33	10
8-3	8	8-98	11
8-4	9	---	---

Shell size 10, figure 3			
Insert arrangement	Page	Insert arrangement	Page
10-6	12	10-98	13

Shell size 12, figure 4			
Insert arrangement	Page	Insert arrangement	Page
12-3	14	12-10	16
12-8	15	---	---

Shell size 14, figure 5			
Insert arrangement	Page	Insert arrangement	Page
14-4	17	14-15	21
14-5	18	14-18	22
14-9	19	14-19	23
14-12	20	14-22	24

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Table III. Cross-reference of insert arrangements by shell size.

Shell size 16, figure 6			
Insert arrangement	Page	Insert arrangement	Page
16-8	25	16-26	28
16-14	26	16-95	29
16-23	27	16-99	30

Shell size 18, figure 7			
Insert arrangement	Page	Insert arrangement	Page
18-8	31	18-32	34
18-11	32	18-85	35
18-30	33	18-88	36

Shell size 20, figure 8			
Insert arrangement	Page	Insert arrangement	Page
20-16	37	20-39	40
20-24	38	20-41	42
20-27	39	20-90	44

Shell size 22, figure 9			
Insert arrangement	Page	Insert arrangement	Page
22-12	45	22-41	51
22-19	46	22-55	53
22-21	47	22-95	55
22-32	48	22-96	57
22-34	49	---	---

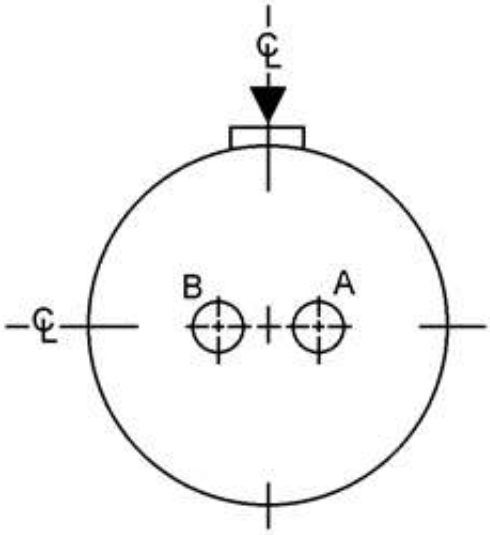
Shell size 24, figure 10			
Insert arrangement	Page	Insert arrangement	Page
24-19	58	24-31	60
24-27	59	24-61	61

6.5 Subject term (key word) listing.

Shell size
Contacts
Socket
Pin

6.6 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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8-2

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.059 (1.50)	.000 (0.00)
B	-.059 (1.50)	.000 (0.00)

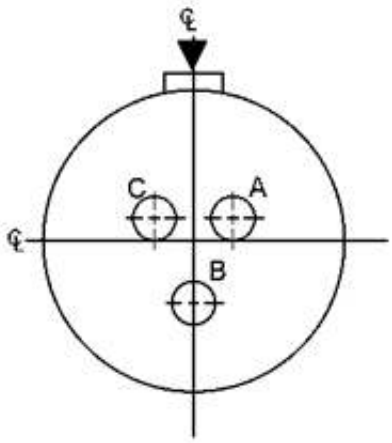
Alternate insert positions			
A degrees			
W	X	Y	Z
58	122	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
8	2	2	20	I	All	MS33708-2

NOTE: Inactive for new design after 11 August 1967.

FIGURE 2. Shell size 8.

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8-3

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.056 (1.42)	+.018 (0.46)
B	-.079 (2.01)	.000 (0.00)
C	-.056 (1.42)	+.018 (0.46)

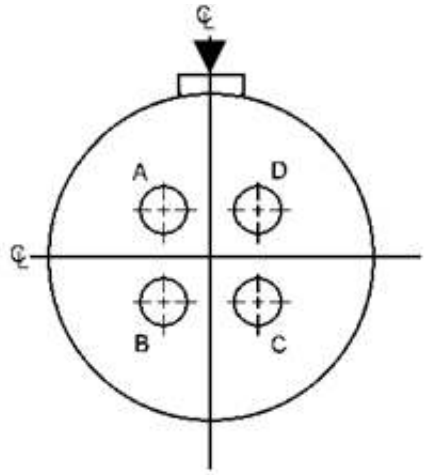
Alternate insert positions			
A degrees			
W	X	Y	Z
60	210	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
8	3	3	20	I	All	MS33708-3

NOTE: Inactive for new design after 1 November 1965.

FIGURE 2. Shell size 8 – Continued.

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8-4

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	-.056 (1.42)	+.056 (1.42)
B	-.056 (1.42)	-.056 (1.42)
C	+.056 (1.42)	-.056 (1.42)
D	+.056 (1.42)	+.056 (1.42)

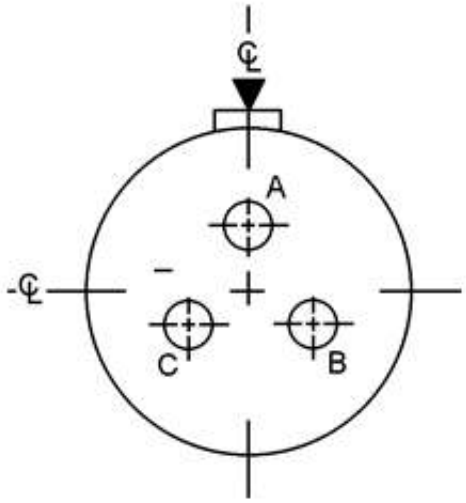
Alternate insert positions			
A degrees			
W	X	Y	Z
45	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
8	4	4	20	I	All	MS33708-4

NOTE: Inactive for new design after 1 November 1965.

FIGURE 2. Shell size 8 – Continued.

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8-33

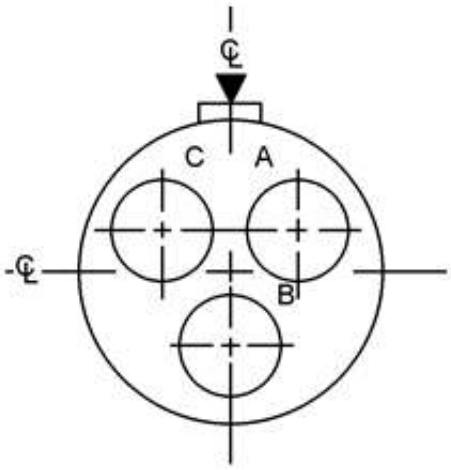
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.075 (1.90)
B	+.065 (1.65)	-.038 (0.97)
C	-.065 (1.65)	-.038 (0.97)

Alternate insert positions			
A degrees			
W	X	Y	Z
90	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
8	33	3	20	I	All	MS33708-33

FIGURE 2. Shell size 8 – Continued.

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8-98

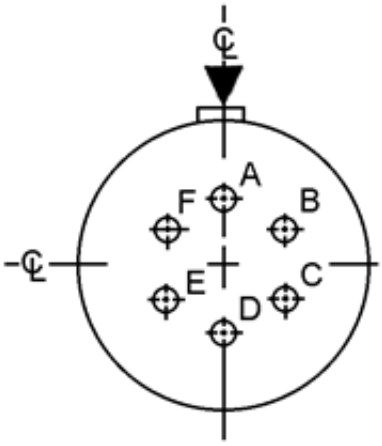
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.037 (0.94)
B	.000 (0.00)	-.075 (1.90)
C	-.065 (1.65)	+.037 (0.94)

Alternate insert positions			
A degrees			
W	X	Y	Z
-----	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
8	98	3	20	I	All	MS33701-98

FIGURE 2. Shell size 8 – Continued.

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10-6

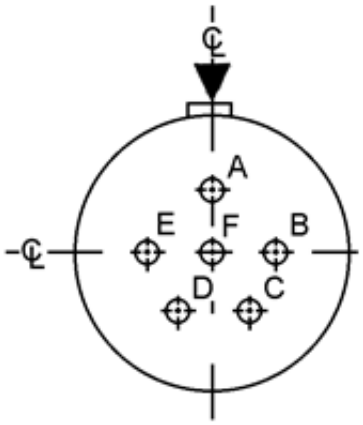
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.130 (3.30)
B	+.113 (2.87)	+.065 (1.65)
C	+.113 (2.87)	-.065 (1.65)
D	.000 (0.00)	-.130 (3.30)
E	-.113 (2.87)	-.065 (1.65)
F	-.113 (2.87)	+.065 (1.65)

Alternate insert positions			
A degrees			
W	X	Y	Z
90	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
10	6	6	20	I	All	MS33709-6

FIGURE 3. Shell size 10.

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10-98

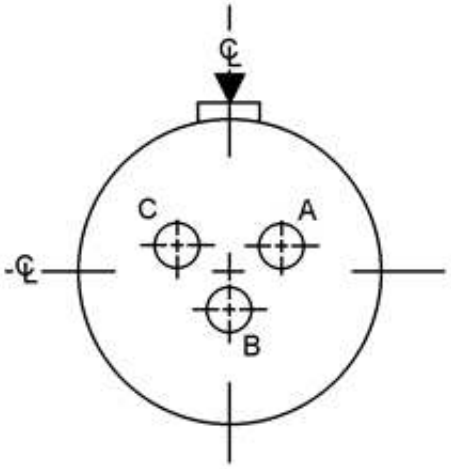
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.130 (3.30)
B	+.130 (3.30)	.000 (0.00)
C	+.065 (1.65)	-.113 (2.87)
D	-.065 (1.65)	-.113 (2.87)
E	-.130 (3.30)	.000 (0.00)
F	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
90	180	240	270

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
10	98	6	20	I	All	MS33709-98

FIGURE 3. Shell size 10 - Continued.

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12-3

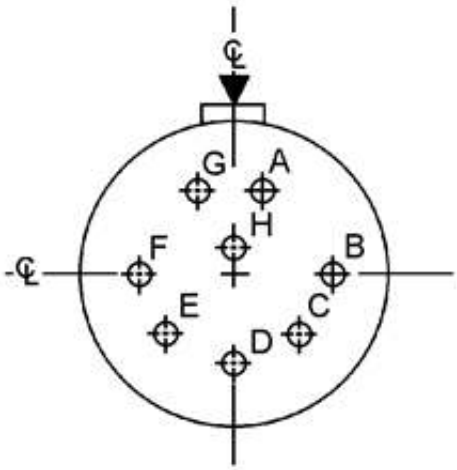
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.094 (2.39)	+.058 (1.47)
B	.000 (0.00)	-.111 (2.82)
C	-.094 (2.39)	+.058 (1.47)

Alternate insert positions			
A degrees			
W	X	Y	Z
-----	-----	180	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
12	3	3	16	II	All	MS33710-3

FIGURE 4. Shell size 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



12-8

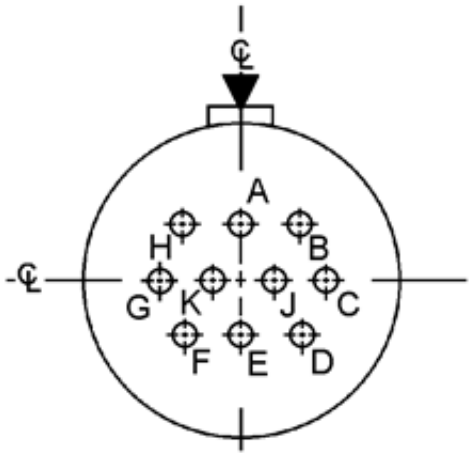
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.157 (3.99)
B	+.170 (4.32)	.000 (0.00)
C	+.120 (3.05)	-.120 (3.05)
D	.000 (0.00)	-.170 (4.32)
E	-.120 (3.05)	-.120 (3.05)
F	-.170 (4.32)	.000 (0.00)
G	-.065 (1.65)	+.157 (3.99)
H	.000 (0.00)	+.044 (1.12)

Alternate insert positions			
A degrees			
W	X	Y	Z
90	112	203	292

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
12	8	8	20	I	All	MS33710-8 MS27727-8

FIGURE 4. Shell size 12 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



12-10

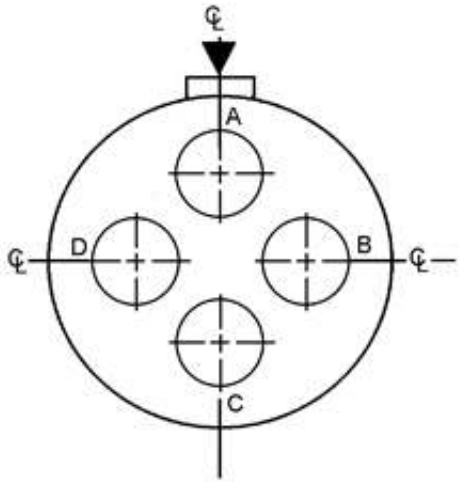
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.120 (3.05)
B	+.130 (3.30)	+.120 (3.05)
C	+.195 (4.95)	.000 (0.00)
D	+.130 (3.30)	-.120 (3.05)
E	.000 (0.00)	-.120 (3.05)
F	-.130 (3.30)	-.120 (3.05)
G	-.195 (4.95)	.000 (0.00)
H	-.130 (3.30)	+.120 (3.05)
J	+.065 (1.65)	.000 (0.00)
K	-.065 (1.65)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
60	155	270	295

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
12	10	10	20	I	All	MS33710-10

FIGURE 4. Shell size 12 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-4

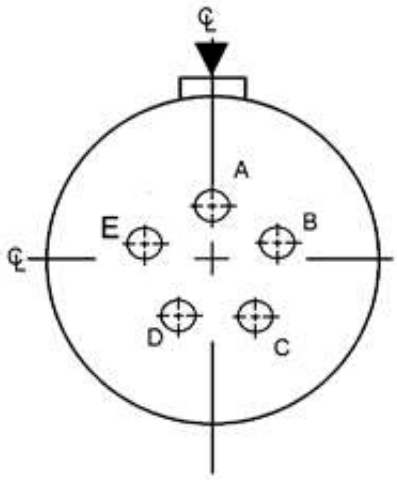
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.169 (4.29)
B	+.169 (4.29)	.000 (0.00)
C	.000 (0.00)	-.169 (4.29)
D	-.169 (4.29)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
45	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	4	4	12 Power	I	All	MS33711-4

FIGURE 5. Shell size 14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-5

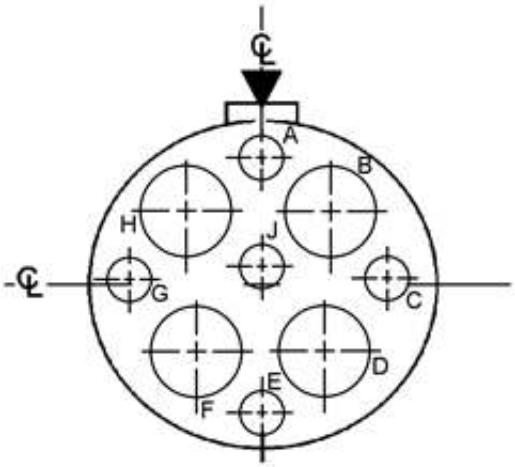
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.100 (2.54)
B	+.174 (4.42)	+.024 (0.61)
C	+.094 (2.39)	-.148 (3.76)
D	-.094 (2.39)	-.148 (3.76)
E	-.174 (4.42)	+.024 (0.61)

Alternate insert positions			
A degrees			
W	X	Y	Z
40	92	184	273

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	5	5	16	II	All	MS33711-5

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-9

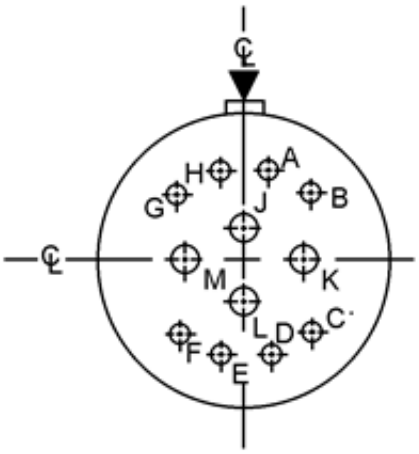
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.260 (6.60)
B	+.141 (3.58)	+.151 (3.83)
C	+.255 (6.48)	+.014 (0.36)
D	+.125 (3.18)	-.124 (3.15)
E	.000 (0.00)	-.253 (6.43)
F	-.125 (3.18)	-.124 (3.15)
G	-.255 (6.48)	+.014 (0.36)
H	-.141 (3.58)	+.151 (3.83)
J	.000 (0.00)	+.042 (1.07)

Alternate insert positions			
A degrees			
W	X	Y	Z
15	90	180	240

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	9	5	20	I	A, C, E, G, J	MS33711-9
		4	12 power		B, D, F, H	

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-12

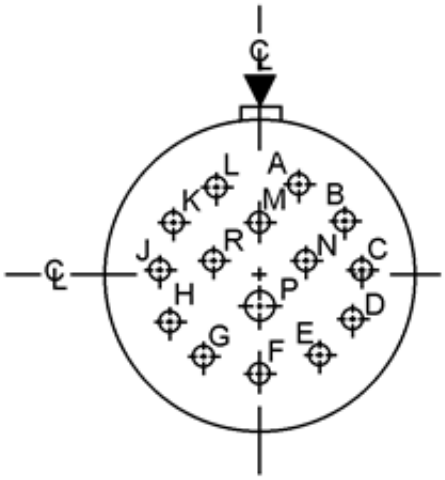
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.243 (6.17)
B	+.178 (4.52)	+.178 (4.52)
C	+.178 (4.52)	-.178 (4.52)
D	+.065 (1.65)	-.243 (6.17)
E	-.065 (1.65)	-.243 (6.17)
F	-.178 (4.52)	-.178 (4.52)
G	-.178 (4.52)	+.178 (4.52)
H	-.065 (1.65)	+.243 (6.17)
J	.000 (0.00)	+.094 (2.39)
K	+.162 (4.11)	.000 (0.00)
L	.000 (0.00)	-.094 (2.39)
M	-.162 (4.11)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
43	90	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	12	4	16	I	J, K, L, M	MS33711-12
		8	20		Remainder	

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-15

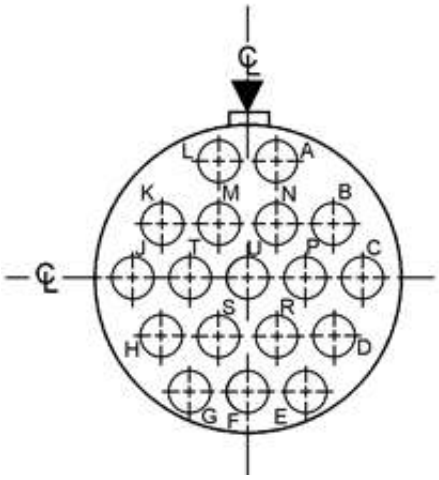
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.225 (5.71)
B	+.202 (5.13)	+.140 (3.56)
C	+.244 (6.20)	+.014 (0.36)
D	+.218 (5.54)	-.113 (2.87)
E	+.126 (3.20)	-.209 (5.31)
F	.000 (0.00)	-.245 (6.22)
G	-.126 (3.20)	-.209 (5.31)
H	-.218 (5.54)	-.113 (2.87)
J	-.244 (6.20)	+.014 (0.36)
K	-.202 (5.13)	+.140 (3.56)
L	-.100 (2.54)	+.225 (5.71)
M	.000 (0.00)	+.140 (3.56)
N	+.110 (2.79)	+.040 (1.02)
P	.000 (0.00)	-.077 (1.96)
R	-.110 (2.79)	+.040 (1.02)

Alternate insert positions			
A degrees			
W	X	Y	Z
17	110	155	234

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	15	1	16	I	P	MS33711-15
		14	20		Remainder	

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-18

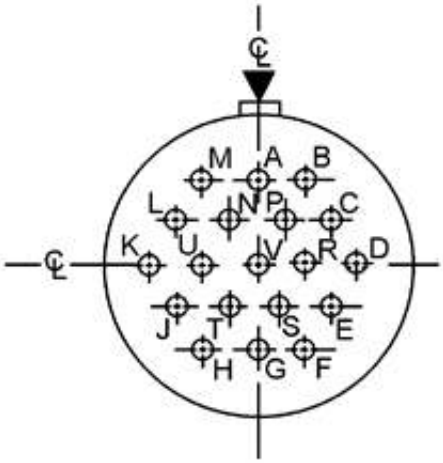
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.252 (6.40)
B	+.195 (4.95)	+.113 (2.87)
C	+.260 (6.60)	.000 (0.00)
D	+.195 (4.95)	-.113 (2.87)
E	+.130 (3.30)	-.225 (5.71)
F	.000 (0.00)	-.225 (5.71)
G	-.130 (3.30)	-.225 (5.71)
H	-.195 (4.95)	-.113 (2.87)
J	-.260 (6.60)	.000 (0.00)
K	-.195 (4.95)	+.113 (2.87)
L	-.065 (1.65)	+.252 (6.40)
M	-.065 (1.65)	+.113 (2.87)
N	+.065 (1.65)	+.113 (2.87)
P	+.130 (3.30)	.000 (0.00)
R	+.065 (1.65)	-.113 (2.87)
S	-.065 (1.65)	-.113 (2.87)
T	-.130 (3.30)	.000 (0.00)
U	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
15	90	180	270

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	18	18	20	I	All	MS33711-18 MS27728-18

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-19

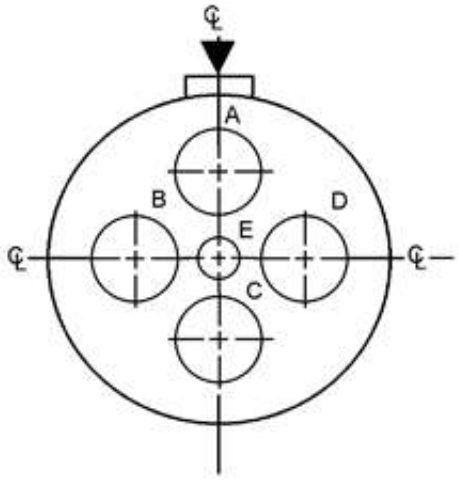
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.225 (5.71)
B	+.130 (3.30)	+.225 (5.71)
C	+.195 (4.95)	+.113 (2.87)
D	+.260 (6.60)	.000 (0.00)
E	+.195 (4.95)	-.113 (2.87)
F	+.130 (3.30)	-.225 (5.71)
G	.000 (0.00)	-.225 (5.71)
H	-.130 (3.30)	-.225 (5.71)
J	-.195 (4.95)	-.113 (2.87)
K	-.260 (6.60)	.000 (0.00)
L	-.195 (4.95)	+.113 (2.87)
M	-.130 (3.30)	+.225 (5.71)
N	-.065 (1.65)	+.113 (2.87)
P	+.065 (1.65)	+.113 (2.87)
R	+.130 (3.30)	.000 (0.00)
S	+.065 (1.65)	-.113 (2.87)
T	-.065 (1.65)	-.113 (2.87)
U	-.130 (3.30)	.000 (0.00)
V	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
30	165	315	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	19	19	20	I	All	MS33711-19

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



14-22

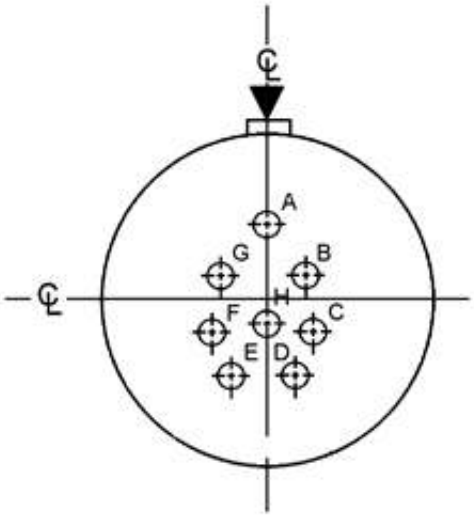
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.191 (4.85)
B	-.191 (4.85)	.000 (0.00)
C	.000 (0.00)	-.191 (4.85)
D	+.191 (4.85)	.000 (0.00)
E	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
45	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
14	22	4	12	I	A, B, C, D	M33711-22
		1	20		E	

FIGURE 5. Shell size 14 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-8

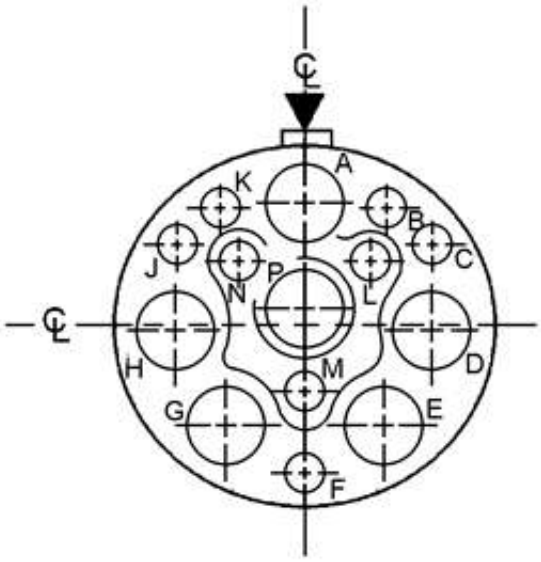
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.236 (5.99)
B	+.128 (3.25)	+.086 (2.18)
C	+.230 (5.60)	-.078 (1.98)
D	+.094 (2.39)	-.216 (5.49)
E	-.094 (2.39)	-.216 (5.49)
F	-.230 (5.60)	-.078 (1.98)
G	-.128 (3.25)	+.086 (2.18)
H	.000 (0.00)	-.052 (1.32)

Alternate insert positions			
A degrees			
W	X	Y	Z
54	152	180	331

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	8	8	16	II	All	MS33712-8

FIGURE 6. Shell size 16.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-14

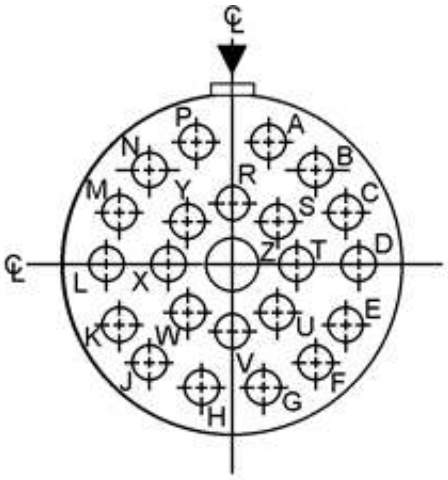
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.268 (6.81)
B	+.178 (4.52)	+.266 (6.76)
C	+.268 (6.81)	+.173 (4.39)
D	+.268 (6.81)	-.005 (.130)
E	+.165 (4.19)	-.212 (5.38)
F	.000 (0.00)	-.320 (8.13)
G	-.165 (4.19)	-.212 (5.38)
H	-.268 (6.81)	-.005 (.130)
J	-.268 (6.81)	+.173 (4.39)
K	-.178 (4.52)	+.266 (6.76)
L	+.142 (3.61)	+.139 (3.53)
M	.000 (0.00)	-.146 (3.71)
N	-.142 (3.61)	+.139 (3.53)
P	.000 (0.00)	+.032 (0.81)

Alternate insert positions			
A degrees			
W	X	Y	Z
25	78	180	240

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	14	8	20	I	B, C, F, J, K, L, M, N	MS33712-14
		6	12		A, D, E, G, H, P	

FIGURE 6. Shell size 16 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-23

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.070 (1.79)	+.300 (7.62)
B	+.193 (4.90)	+.240 (6.10)
C	+.278 (7.06)	+.134 (3.40)
D	+.308 (7.82)	.000 (0.00)
E	+.278 (7.06)	-.134 (3.40)
F	+.193 (4.90)	-.240 (6.10)
G	+.070 (1.79)	-.300 (7.62)
H	-.070 (1.79)	-.300 (7.62)
J	-.193 (4.90)	-.240 (6.10)
K	-.278 (7.06)	-.134 (3.40)
L	-.308 (7.82)	.000 (0.00)
M	-.278 (7.06)	+.134 (3.40)
N	-.193 (4.90)	+.240 (6.10)
P	-.070 (1.79)	+.300 (7.62)
R	.000 (0.00)	+.178 (4.52)
S	+.126 (3.20)	+.126 (3.20)

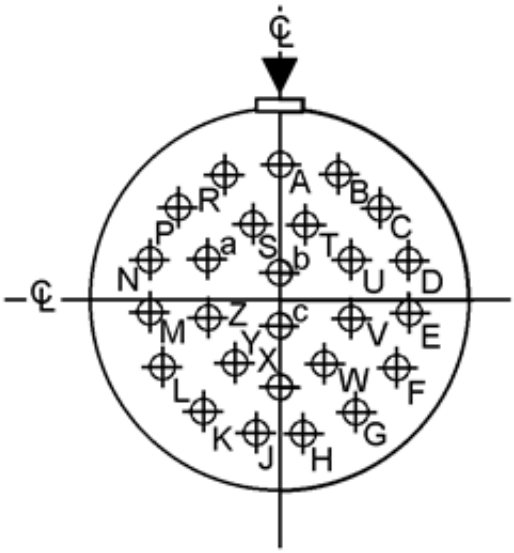
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
T	+.178 (4.52)	.000 (0.00)
U	+.126 (3.20)	-.126 (3.20)
V	.000 (0.00)	-.178 (4.52)
W	-.126 (3.20)	-.126 (3.20)
X	-.178 (4.52)	.000 (0.00)
Y	-.126 (3.20)	+.126 (3.20)
Z	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
158	270	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	23	1	16	I	Z	MS33712-23
		22	20		Remainder	MS33729-23

FIGURE 6. Shell size 16 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-26

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.321 (8.15)
B	+.131 (3.33)	+.293 (7.44)
C	+.239 (6.07)	+.214 (5.44)
D	+.305 (7.75)	+.099 (2.51)
E	+.319 (8.10)	-.034 (0.86)
F	+.278 (7.06)	-.161 (4.09)
G	+.189 (4.80)	-.260 (6.60)
H	+.067 (1.70)	-.314 (7.98)
J	-.067 (1.70)	-.314 (7.98)
K	-.189 (4.80)	-.260 (6.60)
L	-.278 (7.06)	-.161 (4.09)
M	-.319 (8.10)	-.034 (0.86)
N	-.305 (7.75)	+.099 (2.51)
P	-.239 (6.07)	+.214 (5.44)
R	-.131 (3.33)	+.293 (7.44)
S	-.070 (1.79)	+.177 (4.50)

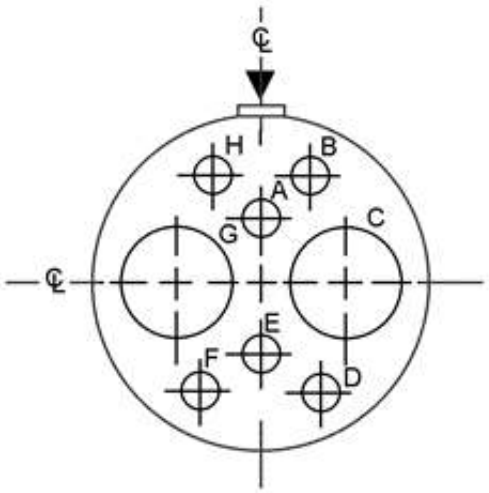
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
T	+.070 (1.79)	+.177 (4.50)
U	+.175 (4.44)	+.094 (2.39)
V	+.178 (4.52)	-.036 (0.91)
W	+.119 (3.02)	-.151 (3.83)
X	.000 (0.00)	-.203 (5.16)
Y	-.119 (3.02)	-.151 (3.83)
Z	-.178 (4.52)	-.036 (0.91)
a	-.175 (4.44)	+.094 (2.39)
b	.000 (0.00)	+.065 (1.65)
c	.000 (0.00)	-.065 (1.65)

Alternate insert positions			
A degrees			
W	X	Y	Z
60	-----	275	338

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	26	26	20	I	All	MS33712-26

FIGURE 6. Shell size 16 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-95

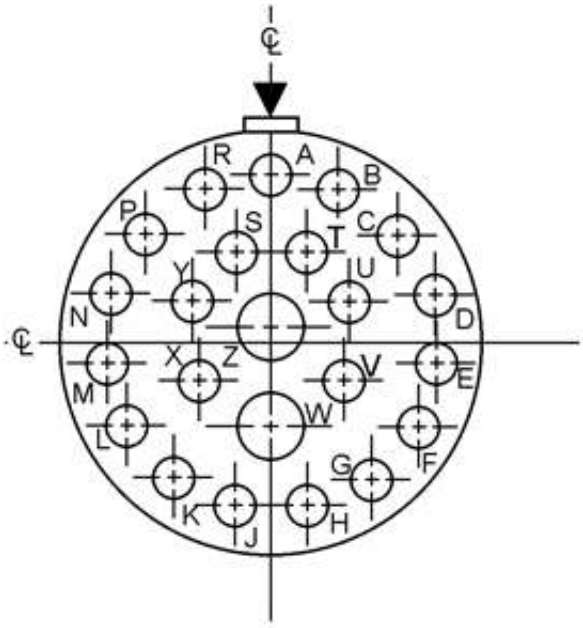
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.162 (4.11)
B	+.115 (2.92)	+.261 (6.83)
C	+.200 (5.08)	.000 (0.00)
D	+.143 (3.63)	-.250 (6.35)
E	.000 (0.00)	-.162 (.411)
F	-.143 (3.63)	-.250 (6.35)
G	-.200 (5.08)	.000 (0.00)
H	-.115 (2.92)	+.261 (6.83)

Alternate insert positions			
A degrees			
W	X	Y	Z
25	90	153	----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	95	6	20	I	A, B, D, E, F, H	MS33712-95
		2	8 shielded. Not applicable to series 2 (rear release connectors)		C, G	

FIGURE 6. Shell size 16 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



16-99

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.321 (8.15)
B	+.131 (3.33)	+.293 (7.44)
C	+.239 (6.07)	+.214 (5.44)
D	+.305 (7.75)	+.099 (2.51)
E	+.319 (8.10)	-.034 (0.86)
F	+.278 (7.06)	-.161 (4.09)
G	+.189 (4.80)	-.260 (6.60)
H	+.067 (1.70)	-.314 (7.98)
J	-.067 (1.70)	-.314 (7.98)
K	-.189 (4.80)	-.260 (6.60)
L	-.278 (7.06)	-.161 (4.09)
M	-.319 (8.10)	-.034 (0.86)
N	-.305 (7.75)	+.099 (2.51)
P	-.239 (6.07)	+.214 (5.44)

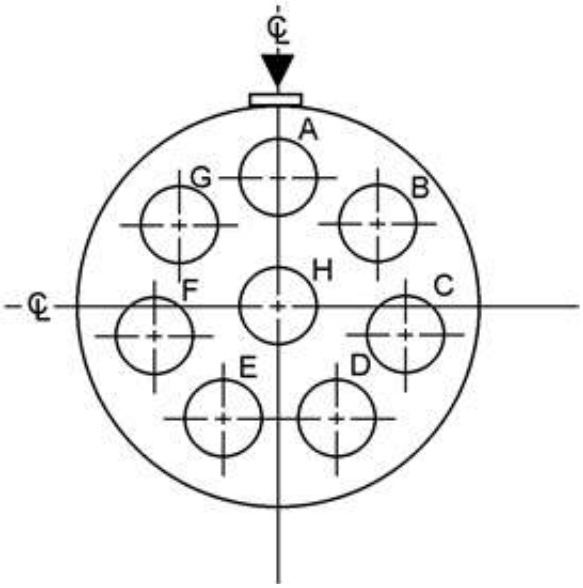
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
R	-.131 (3.33)	+.293 (7.44)
S	-.070 (1.78)	+.177 (4.50)
T	+.070 (1.78)	+.177 (4.50)
U	+.175 (4.44)	+.094 (2.39)
V	+.150 (3.81)	-.075 (1.90)
W	.000 (0.00)	-.161 (4.09)
X	-.150 (3.81)	-.075 (1.90)
Y	-.175 (4.44)	+.094 (2.39)
Z	.000 (0.00)	+.025 (0.63)

Alternate insert positions			
A degrees			
W	X	Y	Z
66	156	223	340

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
16	99	2	16	I	W, Z	MS33712-99
		21	20		Remainder	

FIGURE 6. Shell size 16 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-8

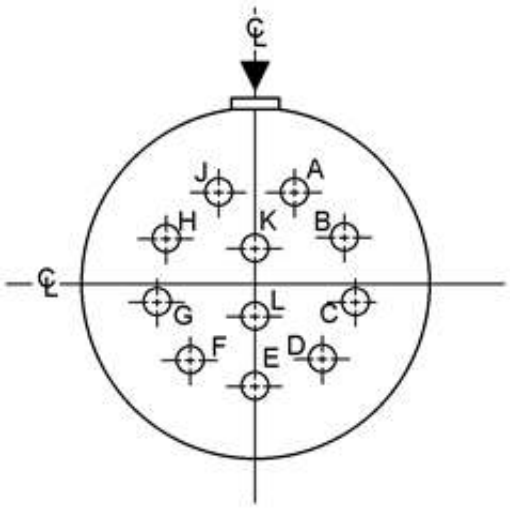
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.280 (7.11)
B	+.219 (5.56)	+.175 (4.45)
C	+.273 (6.93)	-.062 (1.57)
D	+.121 (3.07)	-.252 (6.40)
E	-.121 (3.07)	-.252 (6.40)
F	-.273 (6.93)	-.062 (1.57)
G	-.219 (5.56)	+.175 (4.45)
H	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
180	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	8	8	12	I	All	MS33713-8 MS27730-8

FIGURE 7. Shell size 18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-11

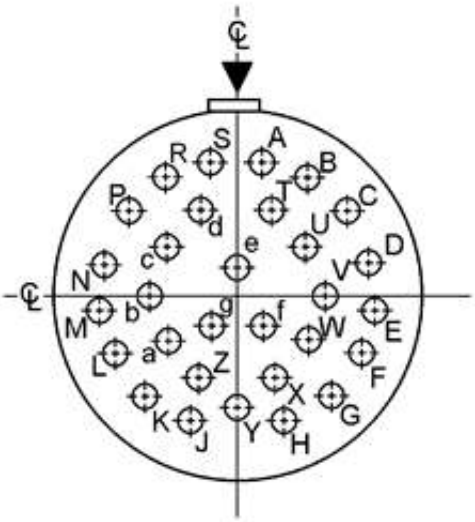
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.105 (2.67)	+.260 (6.60)
B	+.250 (6.35)	+.132 (3.35)
C	+.275 (6.99)	-.053 (1.35)
D	+.179 (4.55)	-.215 (5.46)
E	.000 (.000)	-.281 (7.14)
F	-.179 (4.55)	-.215 (5.46)
G	-.275 (6.99)	-.053 (1.35)
H	-.250 (6.35)	+.132 (3.35)
J	-.105 (2.67)	+.260 (6.60)
K	.000 (.000)	+.105 (2.67)
L	.000 (.000)	-.092 (2.34)

Alternate insert positions			
A degrees			
W	X	Y	Z
62	119	241	340

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	11	11	16	II	All	MS33713-11

FIGURE 7. Shell size 18 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-30

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.346 (8.79)
B	+.186 (4.72)	+.299 (7.59)
C	+.282 (7.16)	+.210 (5.33)
D	+.340 (8.67)	+.093 (2.36)
E	+.351 (8.91)	-.033 (0.84)
F	+.315 (8.00)	-.158 (4.01)
G	+.236 (5.99)	-.261 (6.63)
H	+.124 (3.15)	-.330 (8.38)
J	-.124 (3.15)	-.330 (8.38)
K	-.236 (5.99)	-.261 (6.63)
L	-.315 (8.00)	-.158 (4.01)
M	-.351 (8.91)	-.033 (0.84)
N	-.340 (8.67)	+.093 (2.36)
P	-.282 (7.16)	+.210 (5.33)
R	-.186 (4.72)	+.299 (7.59)
S	-.065 (1.65)	+.346 (8.79)
T	+.084 (2.13)	+.217 (5.51)
U	+.181 (4.60)	+.129 (3.28)

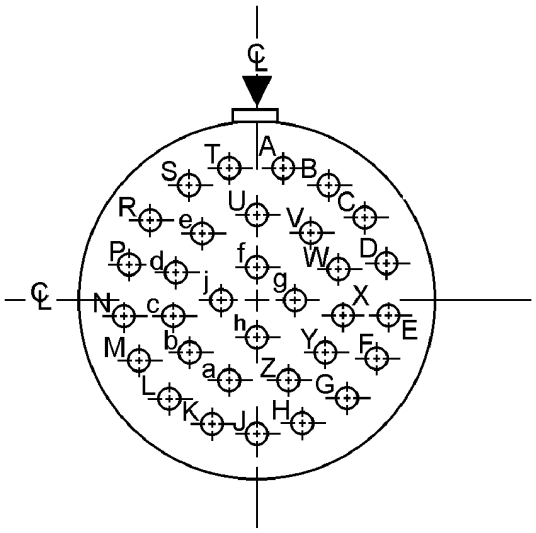
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
V	+.228 (5.79)	+.008 (0.20)
W	+.193 (4.90)	-.117 (2.97)
X	+.096 (2.44)	-.203 (5.16)
Y	.000 (0.00)	-.290 (7.37)
Z	-.096 (2.44)	-.203 (5.16)
a	-.193 (4.90)	-.117 (2.97)
b	-.228 (5.79)	+.008 (0.20)
c	-.181 (4.60)	+.129 (3.28)
d	-.084 (2.13)	+.217 (5.51)
e	.000 (0.00)	+.072 (1.83)
f	+.069 (1.75)	-.076 (1.93)
g	-.069 (1.75)	-.076 (1.93)

Alternate insert positions			
A degrees			
W	X	Y	Z
180	193	285	350

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	30	1	16	I	e	MS33713-30
		29	20		Remainder	MS27730-30

FIGURE 7. Shell size 18 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-32

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.066 (1.68)	+.353 (8.97)
B	+.189 (4.80)	+.305 (7.75)
C	+.286 (7.26)	+.217 (5.51)
D	+.345 (8.76)	+.098 (2.49)
E	+.357 (9.07)	-.033 (0.84)
F	+.321 (8.15)	-.160 (4.06)
G	+.242 (6.15)	-.265 (6.73)
H	+.130 (3.30)	-.335 (8.51)
J	.000 (0.00)	-.359 (9.12)
K	-.130 (3.30)	-.335 (8.51)
L	-.242 (6.15)	-.265 (6.73)
M	-.321 (8.15)	-.160 (4.06)
N	-.357 (9.07)	-.033 (0.84)
P	-.345 (8.76)	+.098 (2.49)
R	-.286 (7.26)	+.217 (5.51)
S	-.189 (4.80)	+.305 (7.75)
T	-.066 (1.68)	+.353 (8.97)
U	.000 (0.00)	+.230 (5.84)
V	+.124 (3.15)	+.193 (4.90)

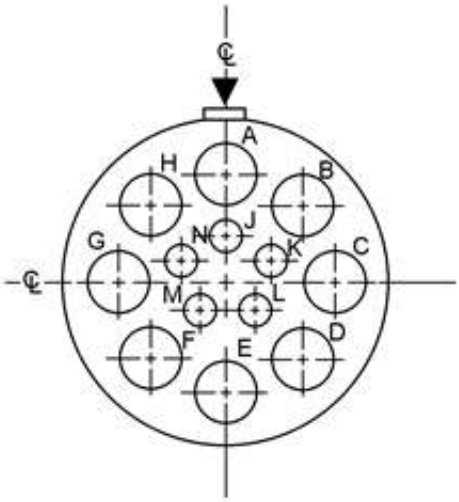
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
W	+.209 (5.31)	+.095 (2.41)
X	+.228 (5.79)	-.033 (0.84)
Y	+.174 (4.42)	-.151 (3.83)
Z	+.065 (1.65)	-.221 (5.61)
a	-.065 (1.65)	-.221 (5.61)
b	-.174 (4.42)	-.151 (3.83)
c	-.228 (5.79)	-.033 (0.84)
d	-.209 (5.31)	+.095 (2.41)
e	-.124 (3.15)	+.193 (4.90)
f	.000 (0.00)	+.096 (2.44)
g	+.096 (2.44)	.000 (0.00)
h	.000 (0.00)	-.096 (2.44)
j	-.096 (2.44)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
85	138	222	265

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	32	32	20	I	All	MS33713-32

FIGURE 7. Shell size 18 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-85

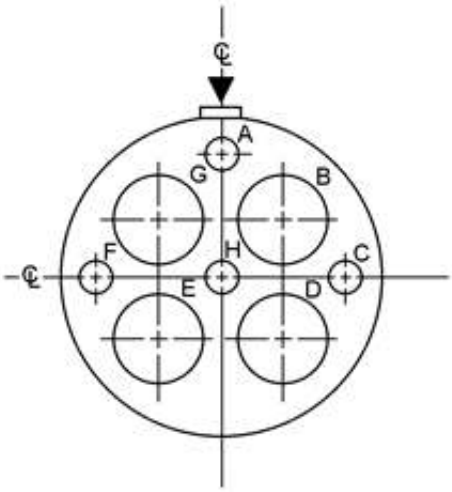
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.305 (7.75)
B	+.215 (5.46)	+.215 (5.46)
C	+.305 (7.75)	.000 (.000)
D	+.215 (5.46)	-.214 (5.44)
E	.000 (.000)	-.300 (7.62)
F	-.215 (5.46)	-.214 (5.44)
G	-.305 (7.75)	.000 (.000)
H	-.215 (5.46)	+.215 (5.46)
J	.000 (.000)	+.122 (3.10)
K	+.126 (3.20)	+.059 (1.50)
L	+.078 (1.98)	-.079 (2.01)
M	-.078 (1.98)	-.079 (2.01)
N	-.126 (3.20)	+.059 (1.50)

Alternate insert positions			
A degrees			
W	X	Y	Z
45	90	180	240

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	85	5	20	I	J, K, L, M, N	MS33713-85
		8	12		Remainder	

FIGURE 7. Shell size 18 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



18-88

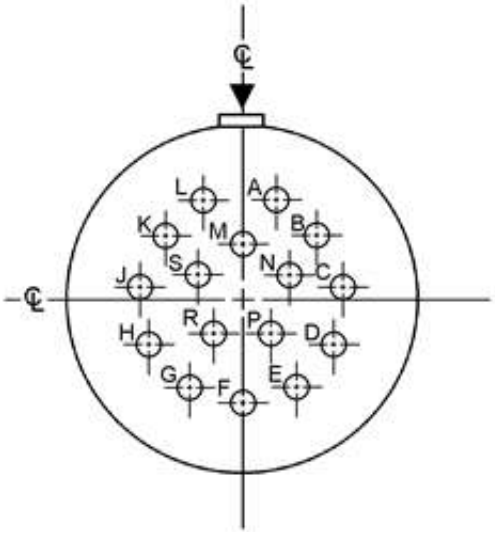
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.350 (8.89)
B	+.175 (4.44)	+.175 (4.44)
C	+.350 (8.89)	.000 (.000)
D	+.175 (4.44)	-.175 (4.44)
E	-.175 (4.44)	-.175 (4.44)
F	-.350 (8.89)	.000 (.000)
G	-.175 (4.44)	+.175 (4.44)
H	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
12	45	168	202

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
18	88	4	20	I	A, C, F, H	MS33713-88
		4	8 shielded. Not applicable to series 2 (rear release connectors).		B, D, E, G	

FIGURE 7. Shell size 18 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



20-16

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.118 (3.00)	+.322 (8.18)
B	+.271 (6.88)	+.211 (5.36)
C	+.341 (8.66)	+.036 (0.91)
D	+.308 (7.82)	-.150 (3.81)
E	+.182 (4.62)	-.290 (7.37)
F	.000 (.000)	-.343 (8.71)
G	-.182 (4.62)	-.290 (7.37)
H	-.308 (7.82)	-.150 (3.81)
J	-.341 (8.66)	+.036 (0.91)
K	-.271 (6.88)	+.211 (5.36)
L	-.118 (3.00)	+.322 (8.18)

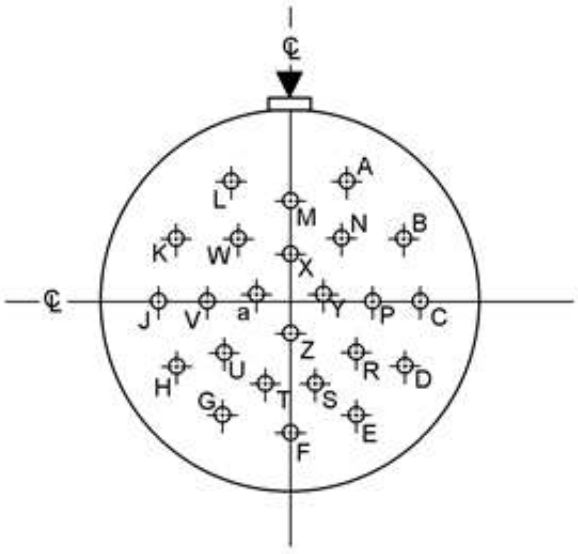
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
M	.000 (.000)	+.175 (4.44)
N	+.154 (3.91)	+.062 (1.57)
P	+.094 (2.39)	-.122 (3.10)
R	-.094 (2.39)	-.122 (3.10)
S	-.154 (3.91)	+.062 (1.57)

Alternate insert positions			
A degrees			
W	X	Y	Z
238	318	333	347

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	16	16	16	II	All	MS33714-16

FIGURE 8. Shell size 20.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



20-24

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.175 (4.44)	+.375 (9.52)
B	+.350 (8.89)	+.200 (5.08)
C	+.400 (10.18)	.000 (.000)
D	+.350 (8.89)	-.200 (5.08)
E	+.200 (5.08)	-.350 (8.89)
F	.000 (.000)	-.400 (10.16)
G	-.200 (5.08)	-.350 (8.89)
H	-.350 (8.89)	-.200 (5.08)
J	-.400 (10.18)	.000 (.000)
K	-.350 (8.89)	+.200 (5.08)
L	-.175 (4.44)	+.375 (9.52)
M	.000 (.000)	+.300 (7.62)
N	+.150 (3.81)	+.200 (5.08)
P	+.250 (6.35)	.000 (.000)
R	+.200 (5.08)	-.150 (3.81)

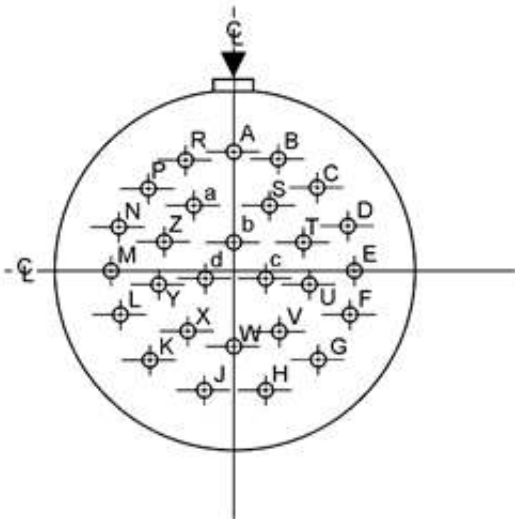
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
S	+.075 (1.90)	-.250 (6.35)
T	-.075 (1.90)	-.250 (6.35)
U	-.200 (5.08)	-.150 (3.81)
V	-.250 (6.35)	.000 (.000)
W	-.150 (3.81)	+.200 (5.08)
X	.000 (.000)	+.150 (3.81)
Y	+.100 (2.54)	+.025 (0.63)
Z	.000 (.000)	-.100 (2.54)
a	-.100 (2.54)	+.025 (0.63)

Alternate insert positions			
A degrees			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	24	24	20	I	All	MS33714-24 MS27731-24

FIGURE 8. Shell size 20 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



20-27

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.400 (10.16)
B	+.150 (3.81)	+.375 (9.53)
C	+.275 (6.99)	+.275 (6.99)
D	+.375 (9.53)	+.150 (3.81)
E	+.400 (10.16)	.000 (.000)
F	+.375 (9.53)	-.150 (3.81)
G	+.275 (6.99)	-.300 (7.62)
H	+.100 (2.54)	-.400 (10.16)
J	-.100 (2.54)	-.400 (10.16)
K	-.275 (6.99)	-.300 (7.62)
L	-.375 (9.53)	-.150 (3.81)
M	-.400 (10.16)	.000 (.000)
N	-.375 (9.53)	+.150 (3.81)
P	-.275 (6.99)	+.275 (6.99)
R	-.150 (3.81)	+.375 (9.53)
S	+.125 (3.18)	+.225 (5.71)
T	+.225 (5.71)	-.100 (2.54)

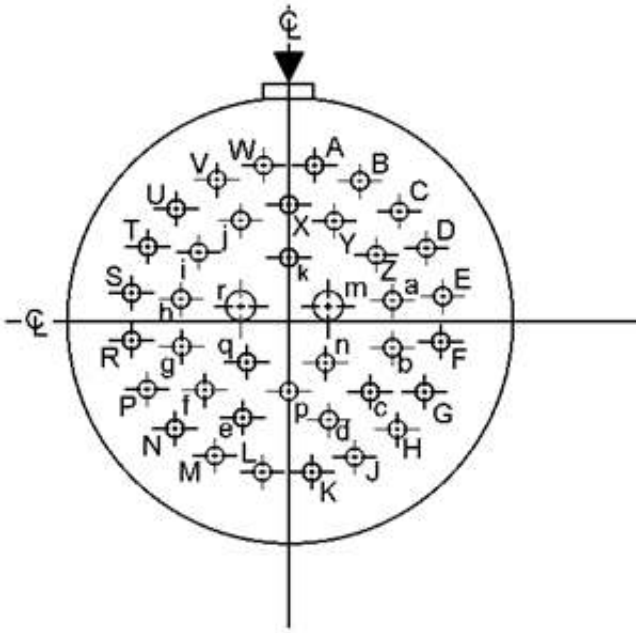
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
U	+.250 (6.35)	-.050 (1.27)
V	+.150 (3.81)	-.200 (5.08)
W	.000 (.000)	-.250 (6.35)
X	-.150 (3.81)	-.200 (5.08)
Y	-.250 (6.35)	-.050 (1.27)
Z	-.225 (5.71)	-.100 (2.54)
a	-.125 (3.18)	+.225 (5.71)
b	.000 (.000)	+.100 (2.54)
c	+.100 (2.54)	-.025 (0.63)
d	-.100 (2.54)	-.025 (0.63)

Alternate insert positions			
A degrees			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	27	27	20	I	All	MS33714-27 MS27731-27

FIGURE 8. Shell size 20 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



20-39

Alternate insert positions			
A degrees			
W	X	Y	Z
63	144	252	333

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	39	2	16	I	m, r	MS33714-39
		37	20		Remainder	MS27731-39

FIGURE 8. Shell size 20 – Continued.

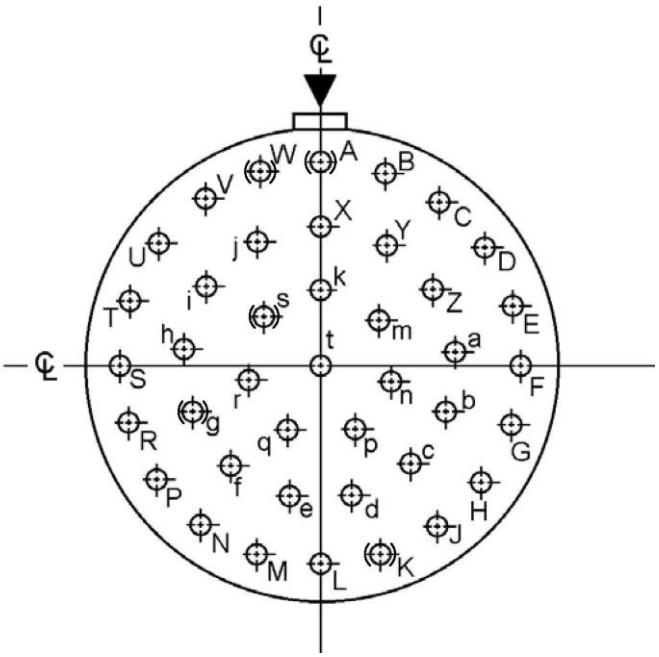
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 39)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.411 (10.44)
B	+.189 (4.80)	+.371 (9.42)
C	+.294 (7.47)	+.294 (7.47)
D	+.371 (9.42)	+.189 (4.80)
E	+.411 (10.44)	+.065 (1.65)
F	+.411 (10.44)	-.065 (1.65)
G	+.371 (9.42)	-.189 (4.80)
H	+.294 (7.47)	-.294 (7.47)
J	+.189 (4.80)	-.371 (9.42)
K	+.065 (1.65)	-.411 (10.44)
L	-.065 (1.65)	-.411 (10.44)
M	-.189 (4.80)	-.371 (9.42)
N	-.294 (7.47)	-.294 (7.47)
P	-.371 (9.42)	-.189 (4.80)
R	-.411 (10.44)	-.065 (1.65)
S	-.411 (10.44)	+.065 (1.65)
T	-.371 (9.42)	+.189 (4.80)
U	-.294 (7.47)	+.294 (7.47)
V	-.189 (4.80)	+.371 (9.42)
W	-.065 (1.65)	+.411 (10.44)

Contacts (arrangement 39)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
X	.000 (.000)	+.295 (7.49)
Y	+.126 (3.20)	+.256 (6.50)
Z	+.232 (5.89)	+.179 (4.55)
a	+.280 (7.11)	+.057 (1.45)
b	+.280 (7.11)	-.074 (1.88)
c	+.217 (5.51)	-.189 (4.80)
d	+.112 (2.84)	-.265 (6.73)
e	-.112 (2.84)	-.265 (6.73)
f	-.217 (5.51)	-.189 (4.80)
g	-.280 (7.11)	-.074 (1.88)
h	-.280 (7.11)	+.057 (1.45)
i	-.232 (5.89)	+.179 (4.55)
j	-.126 (3.20)	+.256 (6.50)
k	.000 (.000)	+.164 (4.17)
m	+.114 (2.90)	+.048 (1.22)
n	+.106 (2.69)	-.107 (2.72)
p	.000 (.000)	+.189 (4.80)
q	-.106 (2.69)	-.107 (2.72)
r	-.114 (2.90)	+.048 (1.22)

FIGURE 8. Shell size 20 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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20-41

Alternate insert positions			
A degrees			
W	X	Y	Z
45	126	225	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	41	41	20	I	All	MS33714-41

FIGURE 8. Shell size 20 – Continued.

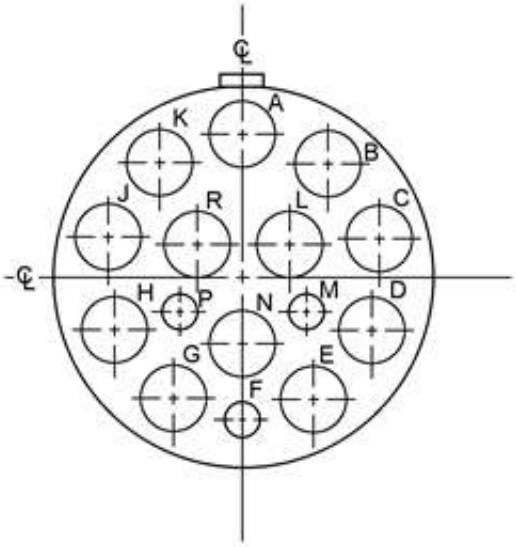
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 41)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.418 (10.62)
B	+.129 (3.28)	+.397 (10.08)
C	+.245 (6.22)	+.338 (8.59)
D	+.338 (8.59)	+.245 (6.22)
E	+.397 (10.08)	+.129 (3.28)
F	+.418 (10.62)	.000 (.000)
G	+.397 (10.08)	-.129 (3.28)
H	+.338 (8.59)	-.245 (6.22)
J	+.245 (6.22)	-.338 (8.59)
K	+.129 (3.28)	-.397 (10.08)
L	.000 (.000)	-.418 (10.62)
M	-.129 (3.28)	-.397 (10.08)
N	-.245 (6.22)	-.338 (8.59)
P	-.338 (8.59)	-.245 (6.22)
R	-.397 (10.08)	-.129 (3.28)
S	-.418 (10.62)	.000 (.000)
T	-.397 (10.08)	+.129 (3.28)
U	-.338 (8.59)	+.245 (6.22)
V	-.245 (6.22)	+.338 (8.59)
W	-.129 (3.28)	+.397 (10.08)
X	.000 (.000)	+.284 (7.20)

Contacts (arrangement 41)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
Y	+.132 (3.35)	+.251 (6.38)
Z	+.233 (5.92)	+.161 (4.09)
a	+.281 (7.14)	+.034 (.867)
b	+.265 (6.73)	-.101 (2.57)
c	+.188 (4.78)	-.212 (5.38)
d	+.068 (1.73)	-.275 (6.99)
e	-.068 (1.73)	-.275 (6.99)
f	-.188 (4.78)	-.212 (5.38)
g	-.265 (6.73)	-.101 (2.57)
h	-.281 (7.14)	+.034 (.867)
i	-.233 (5.92)	+.161 (4.09)
j	-.132 (3.35)	+.251 (6.38)
k	.000 (.000)	+.150 (3.81)
m	+.117 (2.97)	+.094 (2.39)
n	+.146 (3.71)	-.033 (0.84)
p	+.065 (1.65)	-.135 (3.43)
q	-.065 (1.65)	-.135 (3.43)
r	-.146 (3.71)	-.033 (0.84)
s	-.117 (2.97)	+.094 (2.39)
t	.000 (.000)	.000 (.000)

FIGURE 8. Shell size 20 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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20-90

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.368 (9.35)
B	+.224 (5.69)	+.292 (7.42)
C	+.356 (9.04)	+.095 (2.41)
D	+.341 (8.66)	-.142 (3.61)
E	+.182 (4.62)	-.319 (8.10)
F	.000 (.000)	-.368 (9.35)
G	-.182 (4.62)	-.319 (8.10)
H	-.341 (8.66)	-.142 (3.61)
J	-.356 (9.04)	+.095 (2.41)
K	-.224 (5.69)	+.292 (7.42)
L	+.120 (3.05)	+.079 (2.01)

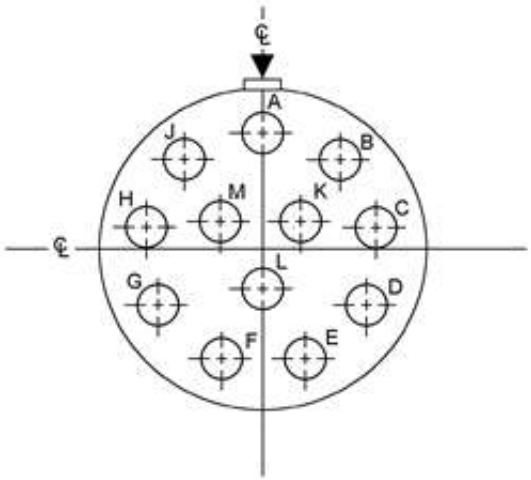
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
M	+.166 (4.22)	-.098 (2.49)
N	.000 (.000)	-.170 (4.32)
P	-.166 (4.22)	-.098 (2.49)
R	-.120 (3.05)	+.079 (2.01)

Alternate insert positions			
A degrees			
W	X	Y	Z
18	60	240	270

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
20	90	3	20	I	F, M, P	MS33714-90
		12	12		Remainder	

FIGURE 8. Shell size 20 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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22-12

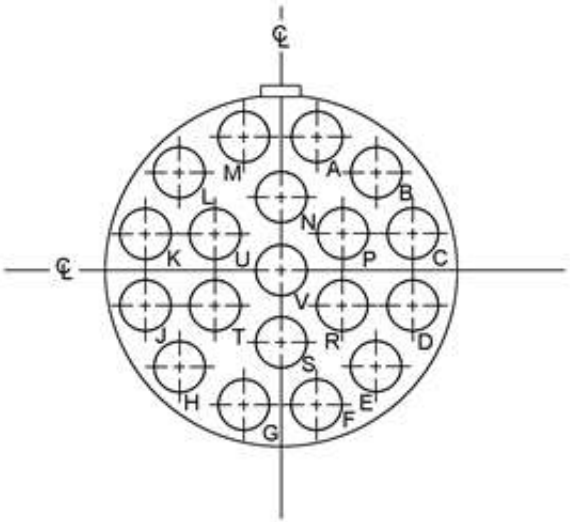
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.398 (10.11)
B	+.256 (6.50)	+.305 (7.75)
C	+.392 (9.97)	+.069 (1.73)
D	+.345 (8.76)	-.199 (5.06)
E	+.136 (3.45)	-.374 (9.50)
F	-.136 (3.45)	-.374 (9.50)
G	-.345 (8.76)	-.199 (5.06)
H	-.392 (9.97)	+.069 (1.73)
J	-.256 (6.50)	+.305 (7.75)
K	+.131 (3.33)	+.075 (1.90)
L	.000 (0.00)	-.151 (3.83)
M	-.131 (3.33)	+.075 (1.90)

Alternate insert positions			
A degrees			
W	X	Y	Z
-----	-----	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	12	12	12	I	All	MS33715-12

FIGURE 9. Shell size 22.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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22-19

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.114 (2.90)	+.425 (10.80)
B	+.311 (7.90)	+.311 (7.90)
C	+.425 (10.80)	+.114 (2.90)
D	+.425 (10.80)	-.114 (2.90)
E	+.311 (7.90)	-.311 (7.90)
F	+.114 (2.90)	-.425 (10.80)
G	-.114 (2.90)	-.425 (10.80)
H	-.311 (7.90)	-.311 (7.90)
J	-.425 (10.80)	-.114 (2.90)
K	-.425 (10.80)	+.114 (2.90)
L	-.311 (7.90)	+.311 (7.90)
M	-.114 (2.90)	+.425 (10.80)
N	.000 (.000)	+.228 (5.79)
P	+.197 (5.00)	+.114 (2.90)
R	+.197 (5.00)	-.114 (2.90)

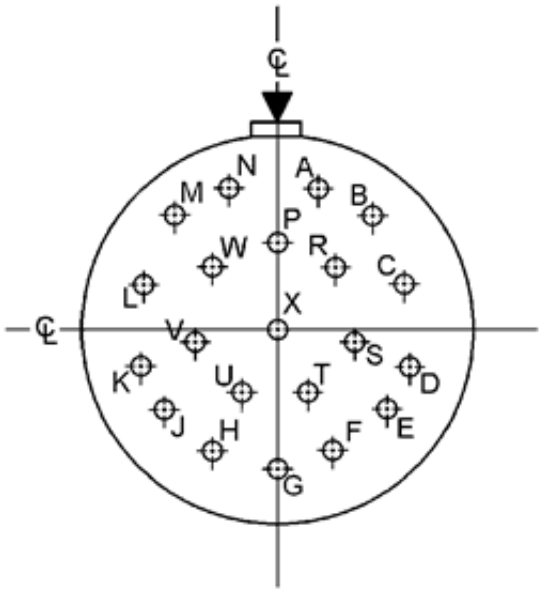
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
S	.000 (.000)	-.228 (5.79)
T	-.197 (5.00)	-.114 (2.90)
U	-.197 (5.00)	+.114 (2.90)
V	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
15	90	225	308

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	19	19	12	I	All	MS33715-19

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



22-21

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.128 (3.25)	+.385 (9.78)
B	+.289 (7.34)	+.285 (7.24)
C	+.386 (9.80)	+.123 (3.12)
D	+.400 (10.16)	-.065 (1.65)
E	+.328 (8.33)	-.239 (6.07)
F	+.183 (4.65)	-.362 (9.19)
G	.000 (.000)	-.406 (10.31)
H	-.183 (4.65)	-.362 (9.19)
J	-.328 (8.33)	-.239 (6.07)
K	-.400 (10.16)	-.065 (1.65)
L	-.386 (9808)	+.123 (3.12)
M	-.289 (7.34)	+.285 (7.24)
N	-.128 (3.25)	+.385 (9.78)
P	.000 (.000)	+.245 (6.22)
R	+.160 (4.06)	+.146 (3.71)

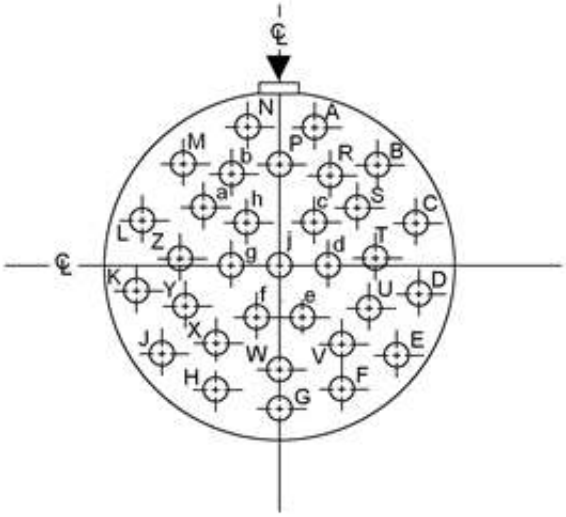
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
S	+.214 (5.44)	-.035 (0.89)
T	+.094 (2.39)	-.194 (4.93)
U	-.094 (2.39)	-.194 (4.93)
V	-.214 (5.44)	-.035 (0.89)
W	-.160 (4.06)	+.146 (3.71)
X	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
16	135	175	349

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	21	21	16	II	All	MS33715-21

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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22-32

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.450 (11.43)
B	+.325 (8.25)	+.325 (8.25)
C	+.425 (10.80)	+.150 (3.81)
D	+.450 (11.43)	-.075 (1.90)
E	+.375 (9.52)	-.275 (6.99)
F	+.200 (5.08)	-.400 (10.16)
G	.000 (.000)	-.450 (11.43)
H	-.200 (5.08)	-.400 (10.16)
J	-.375 (9.52)	-.275 (6.99)
K	-.450 (11.43)	-.075 (1.90)
L	-.425 (10.80)	+.150 (3.81)
M	-.325 (8.25)	+.325 (8.25)
N	-.100 (2.54)	+.450 (11.43)
P	.000 (.000)	+.325 (8.25)
R	+.150 (3.81)	+.300 (7.62)
S	+.250 (6.35)	+.175 (4.44)
T	+.325 (8.25)	+.025 (0.63)
U	+.300 (7.62)	-.125 (3.18)
V	+.200 (5.08)	-.250 (6.35)

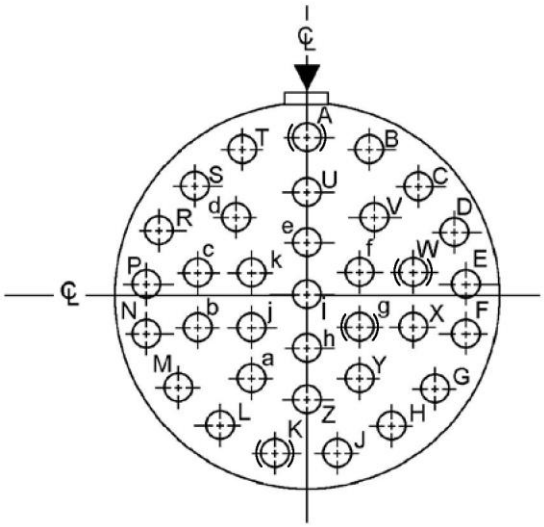
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
W	.000 (.000)	-.300 (7.62)
X	-.200 (5.08)	-.250 (6.35)
Y	-.300 (7.62)	-.125 (3.18)
Z	-.325 (8.25)	+.025 (0.63)
a	-.250 (6.35)	+.175 (4.44)
b	-.150 (3.81)	+.300 (7.62)
c	+.100 (2.54)	+.150 (3.81)
d	+.150 (3.81)	.000 (.000)
e	+.075 (1.90)	-.150 (3.81)
f	-.075 (1.90)	-.150 (3.81)
g	-.150 (3.81)	.000 (.000)
h	-.100 (2.54)	+.150 (3.81)
j	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
72	145	215	288

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	32	32	20	I	All	MS33715-32 MS27732-32 MS33715-18

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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22-34

Alternate insert positions			
A degrees			
W	X	Y	Z
62	142	218	298

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	34	34	20	I	All	MS33715-34 MS27732-34

FIGURE 9. Shell size 22 – Continued.

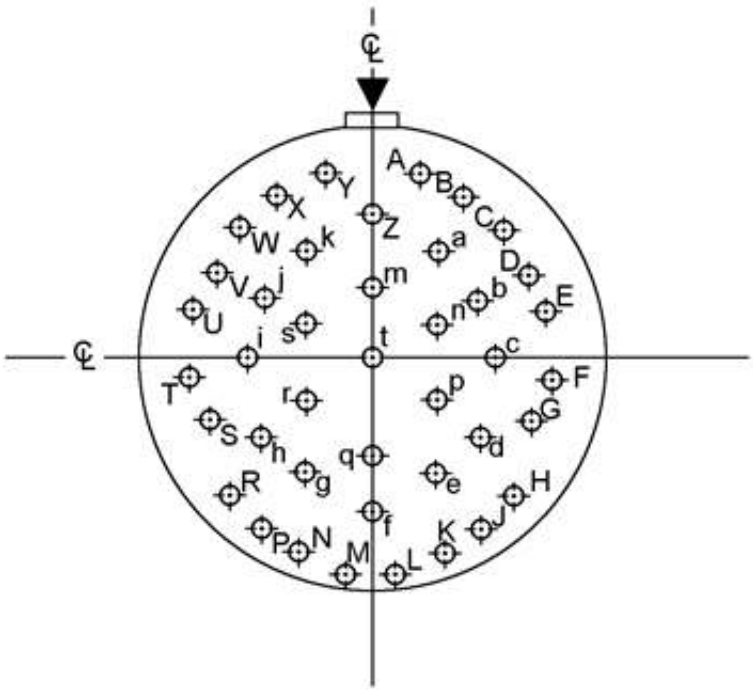
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 34)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.450 (11.43)
B	+.175 (4.44)	+.425 (10.80)
C	+.325 (8.25)	+.325 (8.25)
D	+.425 (10.80)	+.175 (4.44)
E	+.450 (11.43)	+.025 (0.63)
F	+.450 (11.43)	-.125 (3.18)
G	+.375 (9.52)	-.275 (6.99)
H	+.250 (6.35)	-.375 (9.52)
J	+.100 (2.54)	-.450 (11.43)
K	-.100 (2.54)	-.450 (11.43)
L	-.250 (6.35)	-.375 (9.52)
M	-.375 (9.52)	-.275 (6.99)
N	-.450 (11.43)	-.125 (3.18)
P	-.450 (11.43)	+.025 (0.63)
R	-.425 (10.80)	+.175 (4.44)
S	-.325 (8.25)	+.325 (8.25)
T	-.175 (4.44)	+.425 (10.80)

Contacts (arrangement 34)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
U	.000 (.000)	+.300 (7.62)
V	+.200 (5.08)	+.225 (5.71)
W	+.300 (7.62)	+.050 (1.27)
X	+.300 (7.62)	-.100 (2.54)
Y	+.150 (3.81)	-.250 (6.35)
Z	.000 (.000)	-.300 (7.62)
a	-.150 (3.81)	-.250 (6.35)
b	-.300 (7.62)	-.100 (2.54)
c	-.300 (7.62)	+.050 (1.27)
d	-.200 (5.08)	+.225 (5.71)
e	.000 (.000)	+.150 (3.81)
f	+.150 (3.81)	+.050 (1.27)
g	+.150 (3.81)	-.100 (2.54)
h	.000 (.000)	-.150 (3.81)
j	-.150 (3.81)	-.100 (2.54)
k	-.150 (3.81)	+.050 (1.27)
l	.000 (.000)	.000 (.000)

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



22-41

Alternate insert positions			
A degrees			
W	X	Y	Z
39	135	264	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	41	14	16	I	Z, a, c, e, g, i, k, m, n, p, q, r, s, t	MS33715-41
		27	20		Remainder	

FIGURE 9. Shell size 22 – Continued.

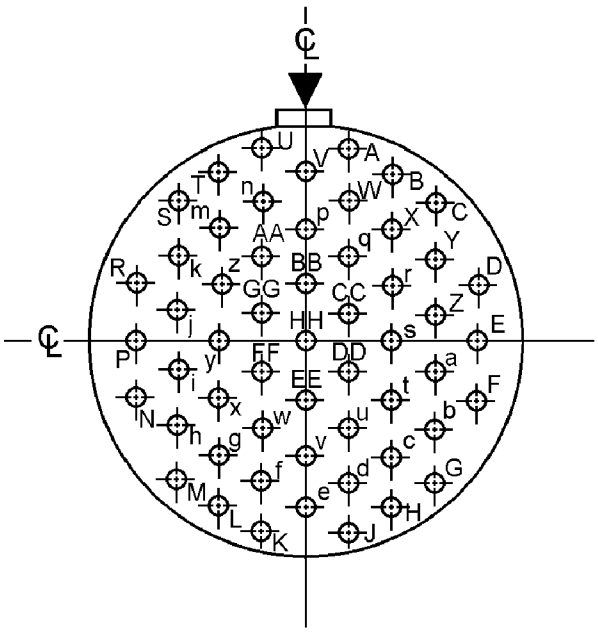
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 41)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.132 (3.35)	+.472 (11.99)
B	+.255 (6.48)	+.418 (10.62)
C	+.358 (9.09)	+.334 (8.48)
D	+.435 (11.04)	+.225 (5.71)
E	+.480 (12.19)	+.099 (2.51)
F	+.489 (12.42)	-.034 (0.86)
G	+.462 (11.73)	-.165 (4.19)
H	+.400 (10.16)	-.283 (7.19)
J	+.308 (7.82)	-.381 (9.68)
K	+.194 (4.93)	-.450 (11.43)
L	+.067 (1.70)	-.486 (12.34)
M	-.067 (1.70)	-.486 (12.34)
N	-.194 (4.93)	-.450 (11.43)
P	-.308 (7.82)	-.381 (9.68)
R	-.400 (10.16)	-.283 (7.19)
S	-.462 (11.73)	-.165 (4.19)
T	-.489 (12.42)	-.034 (0.86)
U	-.480 (12.19)	+.099 (2.51)
V	-.435 (11.04)	+.225 (5.71)
W	-.358 (9.09)	+.334 (8.48)
X	-.255 (6.48)	+.418 (10.62)

Contacts (arrangement 41)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
Y	-.132 (3.35)	+.472 (11.99)
Z	.000 (.000)	+.376 (9.55)
a	+.163 (4.14)	+.282 (7.16)
b	+.299 (7.59)	+.173 (4.39)
c	+.326 (8.28)	.000 (.000)
d	+.308 (7.82)	-.173 (4.39)
e	+.163 (4.14)	-.282 (7.16)
f	.000 (.000)	-.344 (8.74)
g	-.163 (4.14)	-.282 (7.16)
h	-.308 (7.82)	-.173 (4.39)
i	-.326 (8.28)	.000 (.000)
j	-.299 (7.59)	+.173 (4.39)
k	-.163 (4.14)	+.282 (7.16)
m	.000 (.000)	+.188 (4.77)
n	+.163 (4.14)	+.094 (2.39)
p	+.163 (4.14)	-.094 (2.39)
q	.000 (.000)	-.188 (4.77)
r	-.163 (4.14)	-.094 (2.39)
s	-.163 (4.14)	+.094 (2.39)
t	.000 (.000)	.000 (.000)

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



22-55

Alternate insert positions			
A degrees			
W	X	Y	Z
30	142	226	314

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	55	55	20	I	All	MS33715-55

FIGURE 9. Shell size 22 – Continued.

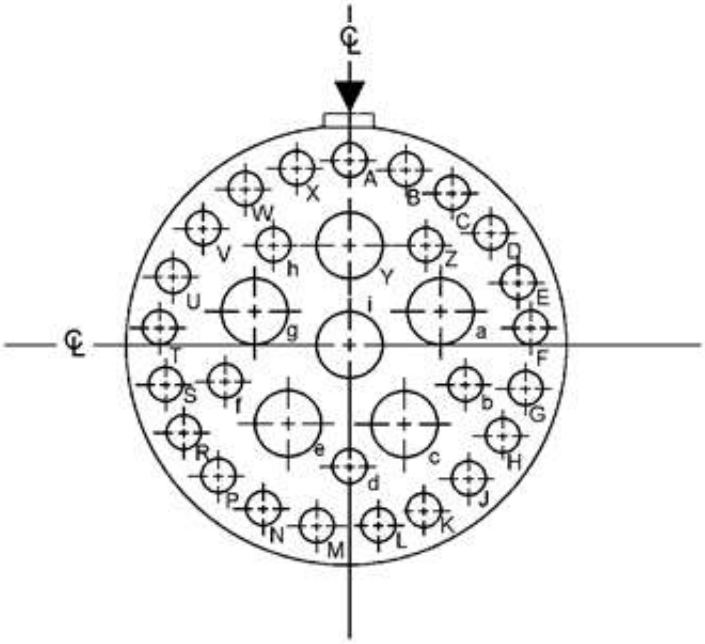
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 55)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.112 (2.84)	+.455 (11.56)
B	+.225 (5.71)	+.390 (9.91)
C	+.336 (8.53)	+.325 (8.25)
D	+.450 (11.43)	+.130 (3.30)
E	+.450 (11.43)	.000 (.000)
F	+.450 (11.43)	-.130 (3.30)
G	+.336 (8.53)	-.325 (8.25)
H	+.225 (5.71)	-.390 (9.91)
J	+.112 (2.84)	-.455 (11.56)
K	-.112 (2.84)	-.455 (11.56)
L	-.225 (5.71)	-.390 (9.91)
M	-.336 (8.53)	-.325 (8.25)
N	-.450 (11.43)	-.130 (3.30)
P	-.450 (11.43)	.000 (.000)
R	-.450 (11.43)	+.130 (3.30)
S	-.336 (8.53)	+.325 (8.25)
T	-.225 (5.71)	+.390 (9.91)
U	-.112 (2.84)	+.455 (11.56)
V	.000 (.000)	+.390 (9.91)
W	+.112 (2.84)	+.325 (8.25)
X	+.225 (5.71)	+.260 (6.60)
Y	+.336 (8.53)	+.195 (4.95)
Z	+.336 (8.53)	+.065 (1.65)
a	+.336 (8.53)	-.065 (1.65)
b	+.336 (8.53)	-.195 (4.95)
c	+.225 (5.71)	-.260 (6.60)
d	+.112 (2.84)	-.325 (8.25)
e	.000 (.000)	-.390 (9.91)

Contacts (arrangement 55)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
f	-.112 (2.84)	-.325 (8.25)
g	-.225 (5.71)	-.260 (6.60)
h	-.336 (8.53)	-.195 (4.95)
i	-.336 (8.53)	-.065 (1.65)
j	-.336 (8.53)	+.065 (1.65)
k	-.336 (8.53)	+.195 (4.95)
m	-.225 (5.71)	+.260 (6.60)
n	-.112 (2.84)	+.325 (8.25)
p	.000 (.000)	+.260 (6.60)
q	+.112 (2.84)	+.195 (4.95)
r	+.225 (5.71)	+.130 (3.30)
s	+.225 (5.71)	.000 (.000)
t	+.225 (5.71)	-.130 (3.30)
u	+.112 (2.84)	-.195 (4.95)
v	.000 (.000)	-.260 (6.60)
w	-.112 (2.84)	-.195 (4.95)
x	-.225 (5.71)	-.130 (3.30)
y	-.225 (5.71)	.000 (.000)
z	-.225 (5.71)	+.130 (3.30)
AA	-.112 (2.84)	+.195 (4.95)
BB	.000 (.000)	+.130 (3.30)
CC	+.112 (2.84)	+.065 (1.65)
DD	+.112 (2.84)	-.065 (1.65)
EE	.000 (.000)	-.130 (3.30)
FF	-.112 (2.84)	-.065 (1.65)
GG	-.112 (2.84)	+.065 (1.65)
HH	.000 (.000)	.000 (.000)

FIGURE 9. Shell size 22 – Continued.

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22-95

Alternate insert positions			
A degrees			
W	X	Y	Z
26	180	266	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	95	6	12	I	Y, a, c, e, g, i	MS33715-95
		26	20		Remainder	

FIGURE 9. Shell size 22 – Continued.

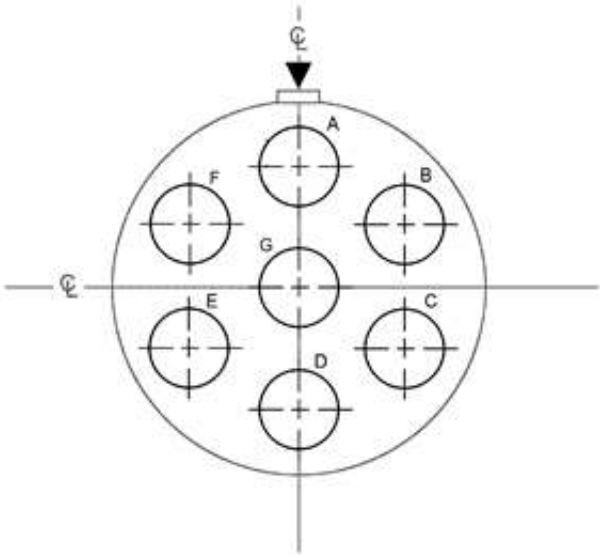
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

Contacts (arrangement 95)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.470 (11.94)
B	+.138 (3.50)	+.449 (11.40)
C	+.265 (6.73)	+.388 (9.86)
D	+.367 (9.32)	+.293 (7.44)
E	+.437 (11.10)	+.172 (4.37)
F	+.469 (11.91)	+.035 (0.89)
G	+.458 (11.63)	-.104 (2.64)
H	+.407 (10.34)	-.234 (5.94)
J	+.320 (8.13)	-.344 (8.74)
K	+.205 (5.21)	-.423 (10.74)
L	+.070 (1.78)	-.465 (11.81)
M	-.070 (1.78)	-.465 (11.81)
N	-.205 (5.21)	-.423 (10.74)
P	-.320 (8.13)	-.344 (8.74)
R	-.407 (10.34)	-.234 (5.94)
S	-.458 (11.63)	-.104 (2.64)

Contacts (arrangement 95)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
T	-.469 (11.91)	+.035 (0.89)
U	-.437 (11.10)	+.172 (4.37)
V	-.367 (9.32)	+.293 (7.44)
W	-.265 (6.73)	+.388 (9.86)
X	-.138 (3.50)	+.449 (11.40)
Y	.000 (.000)	+.250 (6.35)
Z	+.191 (4.85)	+.283 (7.19)
a	+.238 (6.04)	+.077 (1.98)
b	+.309 (7.85)	-.100 (2.54)
c	+.147 (3.73)	-.202 (5.13)
d	.000 (.000)	-.325 (8.25)
e	-.147 (3.73)	-.202 (5.13)
f	-.309 (7.85)	-.100 (2.54)
g	-.238 (6.04)	+.077 (1.98)
h	-.191 (4.85)	+.283 (7.19)
i	.000 (.000)	.000 (.000)

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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22-96

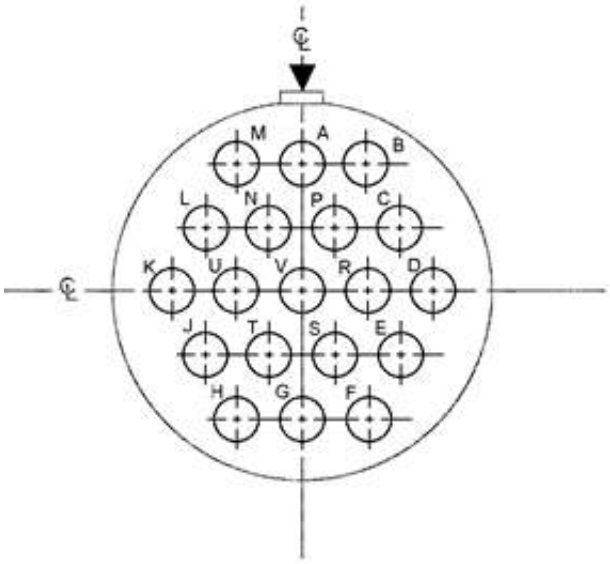
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.360 (9.44)
B	+.312 (7.92)	+.180 (4.57)
C	+.312 (7.92)	-.180 (4.57)
D	.000 (0.00)	-.360 (9.44)
E	-.312 (7.92)	-.180 (4.57)
F	-.312 (7.92)	+.180 (4.57)
G	.000 (0.00)	.000 (0.00)

Alternate insert positions			
A degrees			
W	X	Y	Z
19	41	-----	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
22	96	7	8 Shielded. Not applicable to series 2 (rear release connectors)	I	All	MS33715-96

FIGURE 9. Shell size 22 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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24-19

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.412 (10.46)
B	+.238 (6.04)	+.412 (10.46)
C	+.357 (9.07)	+.206 (5.23)
D	+.476 (12.09)	.000 (.000)
E	+.357 (9.07)	-.206 (5.23)
F	+.238 (6.04)	-.412 (10.46)
G	.000 (.000)	-.412 (10.46)
H	-.238 (6.04)	-.412 (10.46)
J	-.357 (9.07)	-.206 (5.23)
K	-.476 (12.09)	.000 (.000)
L	-.357 (9.07)	+.206 (5.23)
M	-.238 (6.04)	+.412 (10.46)
N	-.119 (3.02)	+.206 (5.23)

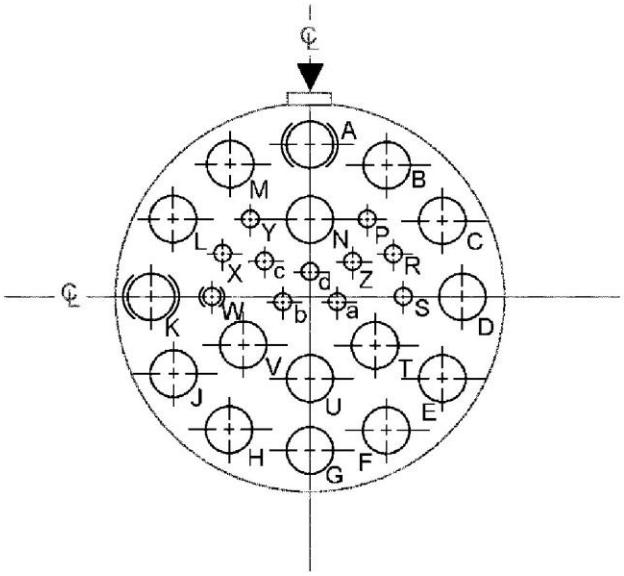
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
P	+.119 (3.02)	+.206 (5.23)
R	+.238 (6.04)	.000 (.000)
S	+.119 (3.02)	-.206 (5.23)
T	-.119 (3.02)	-.206 (5.23)
U	-.238 (6.04)	.000 (.000)
V	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
30	165	315	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
24	19	19	12	II	All	MS33716-19 MS27733-19

FIGURE 10. Shell size 24.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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24-27

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	.000 (.000)	+.491 (12.47)
B	+.246 (6.25)	+.425 (10.80)
C	+.425 (10.80)	+.246 (6.25)
D	+.491 (12.47)	.000 (.000)
E	+.425 (10.80)	-.246 (6.25)
F	+.246 (6.25)	-.425 (10.80)
G	.000 (.000)	-.491 (12.47)
H	-.246 (6.25)	-.425 (10.80)
J	-.425 (10.80)	-.246 (6.25)
K	-.491 (12.47)	.000 (.000)
L	-.425 (10.80)	+.246 (6.25)
M	-.246 (6.25)	+.425 (10.80)
N	.000 (.000)	+.255 (6.48)
P	+.184 (4.67)	+.239 (6.07)
R	+.270 (6.86)	+.130 (3.30)
S	+.301 (7.64)	.000 (.000)

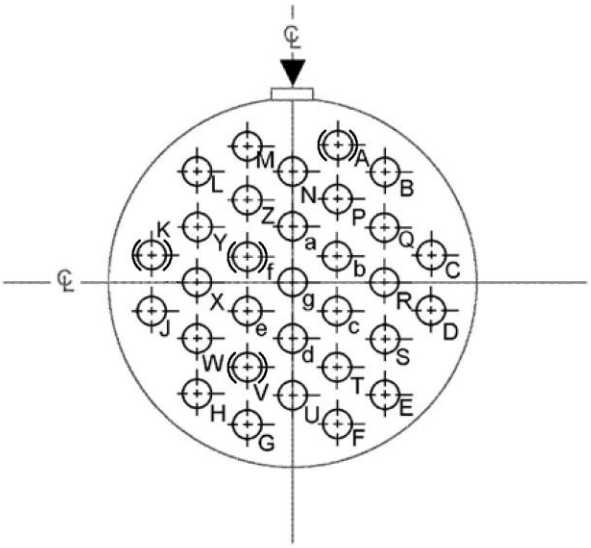
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
T	+.210 (5.33)	-.156 (3.97)
U	.000 (.000)	-.255 (6.48)
V	-.210 (5.33)	-.156 (3.97)
W	-.301 (7.64)	.000 (.000)
X	-.270 (6.86)	+.130 (3.30)
Y	-.184 (4.67)	+.239 (6.07)
Z	+.137 (3.48)	+.103 (2.62)
a	+.088 (2.23)	-.022 (0.56)
b	-.088 (2.23)	-.022 (0.56)
c	-.137 (3.48)	+.103 (2.62)
d	.000 (.000)	+.075 (1.90)

Alternate insert positions			
A degrees			
W	X	Y	Z
45	110	140	225

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
24	27	11	20	I	P, R, S, W, X, Y, Z, a, b, c, d	MS33716-27
		16	12		Remainder	

FIGURE 10. Shell size 24 – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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24-31

Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.158 (4.01)	+.455 (11.56)
B	+.316 (8.03)	+.364 (9.25)
C	+.474 (12.04)	+.091 (2.31)
D	+.474 (12.04)	-.091 (2.31)
E	+.316 (8.03)	-.364 (9.25)
F	+.158 (4.01)	-.455 (11.56)
G	-.158 (4.01)	-.455 (11.56)
H	-.316 (8.03)	-.364 (9.25)
J	-.474 (12.04)	-.091 (2.31)
K	-.474 (12.04)	+.091 (2.31)
L	-.316 (8.03)	+.364 (9.25)
M	-.158 (4.01)	-.455 (11.56)
N	.000 (.000)	+.364 (9.25)
P	+.158 (4.01)	+.273 (6.93)
Q	+.316 (8.03)	+.182 (4.62)
R	+.316 (8.03)	.000 (.000)
S	+.316 (8.03)	-.182 (4.62)
T	+.158 (4.01)	-.273 (6.93)

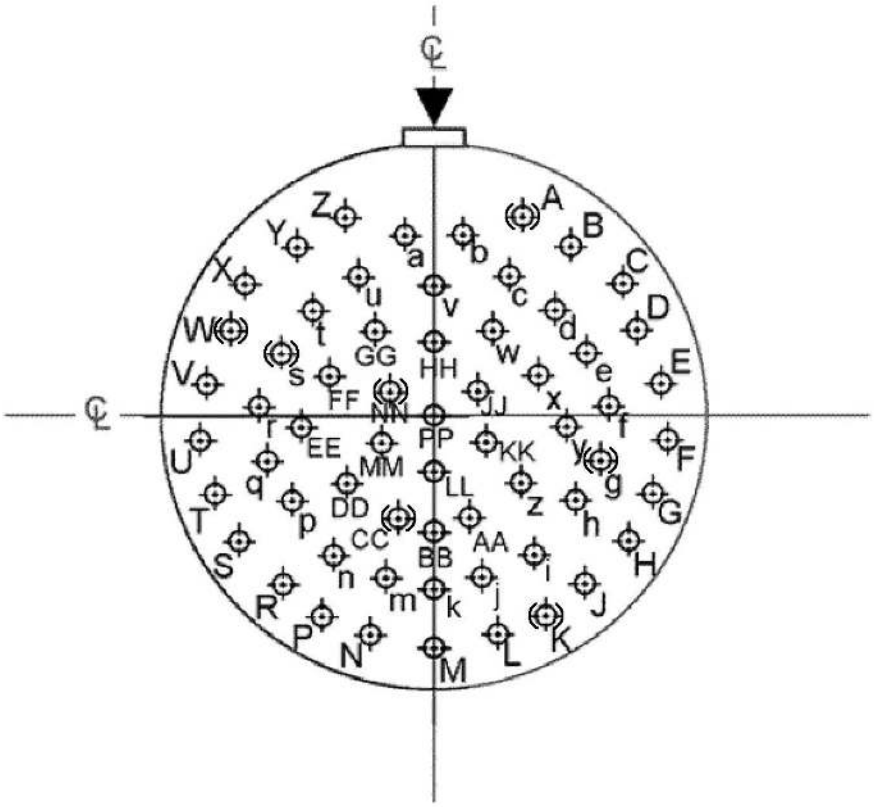
Contacts		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
U	.000 (.000)	-.364 (9.25)
V	-.158 (4.01)	-.273 (6.93)
W	-.316 (8.03)	-.182 (4.62)
X	-.316 (8.03)	.000 (.000)
Y	-.316 (8.03)	+.182 (4.62)
Z	-.158 (4.01)	+.273 (6.93)
a	.000 (.000)	+.182 (4.62)
b	+.158 (4.01)	+.091 (2.31)
c	+.158 (4.01)	-.091 (2.31)
d	.000 (.000)	-.182 (4.62)
e	-.158 (4.01)	-.091 (2.31)
f	-.158 (4.01)	+.091 (2.31)
g	.000 (.000)	.000 (.000)

Alternate insert positions			
A degrees			
W	X	Y	Z
90	225	255	-----

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
24	31	31	16	I	All	MS33716-31 MS27737-31

FIGURE 10. Shell size 24 – Continued.

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24-61

Alternate insert positions			
A degrees			
W	X	Y	Z
90	180	270	324

Shell size	Arrangement number	Number of contacts	Size contacts	Service rating	Contact location	Supersedes document
24	61	61	20	I	All	MS33716-61

FIGURE 10. Shell size 24 – Continued.

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Contacts (arrangement 61)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.196 (4.98)	+.500 (12.70)
B	+.314 (7.98)	+.435 (11.05)
C	+.413 (10.49)	+.343 (8.71)
D	+.485 (12.32)	+.230 (5.84)
E	+.527 (13.39)	+.101 (2.57)
F	+.536 (13.61)	-.030 (0.76)
G	+.511 (12.98)	-.164 (4.17)
H	+.454 (11.53)	-.287 (7.29)
J	+.368 (9.35)	-.391 (9.93)
K	+.259 (6.58)	-.470 (11.94)
L	+.134 (3.40)	-.519 (13.18)
M	.000 (.000)	-.537 (13.64)
N	-.134 (3.40)	-.519 (13.18)
P	-.259 (6.58)	-.470 (11.94)
R	-.368 (9.35)	-.391 (9.93)
S	-.454 (11.53)	-.287 (7.29)
T	-.511 (12.98)	-.164 (4.17)
U	-.536 (13.61)	-.030 (0.76)
V	-.527 (13.39)	+.101 (2.57)
W	-.485 (12.32)	+.230 (5.84)
X	-.413 (10.49)	+.343 (8.71)
Y	-.314 (7.98)	+.435 (11.05)
Z	-.196 (4.98)	+.500 (12.70)
a	-.068 (1.73)	+.454 (11.53)
b	+.068 (1.73)	+.454 (11.53)
c	+.173 (4.39)	+.363 (9.22)
d	+.285 (7.24)	+.283 (7.19)
e	+.362 (9.19)	+.175 (4.44)
f	+.399 (10.13)	+.046 (1.17)
g	+.392 (9.97)	-.088 (2.23)
h	+.341 (8.68)	-.213 (5.41)

Contacts (arrangement 61)		
Contact position	Contact location	
	X-axis (mm)	Y-axis (mm)
i	+.251 (6.37)	-.314 (7.98)
j	+.133 (3.38)	-.379 (9.627)
k	.000 (.000)	-.402 (10.21)
m	-.133 (3.38)	-.379 (9.627)
n	-.251 (6.37)	-.314 (7.98)
p	-.341 (8.68)	-.213 (5.41)
q	-.392 (9.97)	-.088 (2.23)
r	-.399 (10.13)	+.046 (1.17)
s	-.362 (9.19)	+.175 (4.44)
t	-.285 (7.24)	+.283 (7.19)
u	-.173 (4.39)	+.363 (9.22)
v	.000 (.000)	+.338 (8.58)
w	+.147 (3.73)	+.223 (5.66)
x	+.237 (6.02)	+.122 (3.10)
y	+.267 (6.78)	-.010 (0.25)
z	+.228 (5.79)	-.139 (3.53)
AA	+.131 (3.33)	-.233 (5.92)
BB	.000 (.000)	-.267 (6.78)
CC	-.131 (3.33)	-.233 (5.92)
DD	-.228 (5.79)	-.139 (3.53)
EE	-.267 (6.78)	-.010 (0.25)
FF	-.237 (6.02)	+.122 (3.10)
GG	-.147 (3.73)	+.223 (5.66)
HH	.000 (.000)	+.200 (5.08)
JJ	+.105 (2.67)	+.094 (2.39)
KK	+.135 (3.43)	-.041 (1.04)
LL	.000 (.000)	-.132 (3.35)
MM	-.135 (3.43)	-.041 (1.04)
NN	-.105 (2.67)	+.094 (2.39)
PP	.000 (.000)	.000 (.000)

FIGURE 10. Shell size 24 – Continued.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935–2016-147)

Review activities:
Army – AR, AV, MI
Navy – EC, SH

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